

Update – Hardware

JC Vaught

Aug 19, 2025

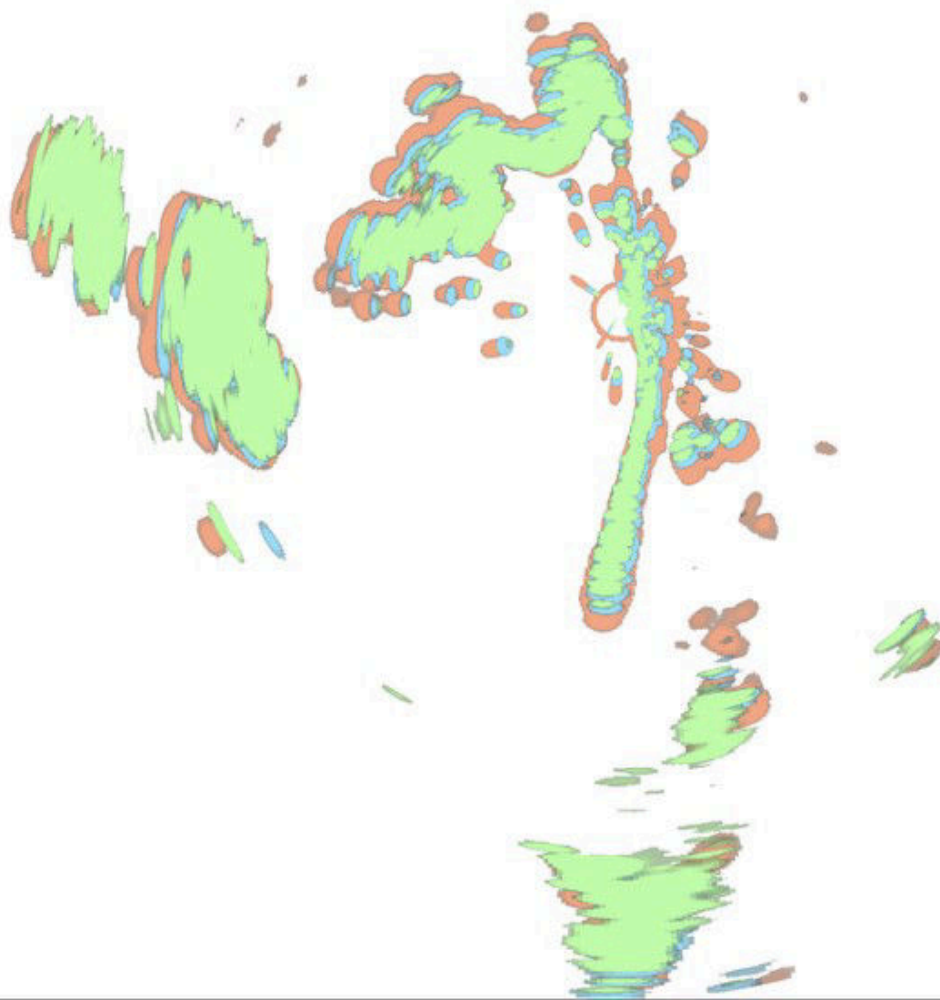
Exec Summary

- Radar
 - Collected more data at lake
 - Created code to collect multi-pulse-length HDR combination
 - Results next week
- AI/ML
 - Tested FFT + Hough TF analysis for object tracking
 - Tried Background subtraction
 - Tested zero shot image segmentation(fail)
 - Built custom MAE implementation 5mil - 2.3b params
 - Tested masking ratios, model size on performance

Radar Overview



Overlay



Object Tracking - Classic

3D FFT SIGNAL PROCESSING

- Temporal sequence 3D data cube creation
- Fast Fourier Transform frequency analysis
- Advanced clutter filtering and spectrum visualization

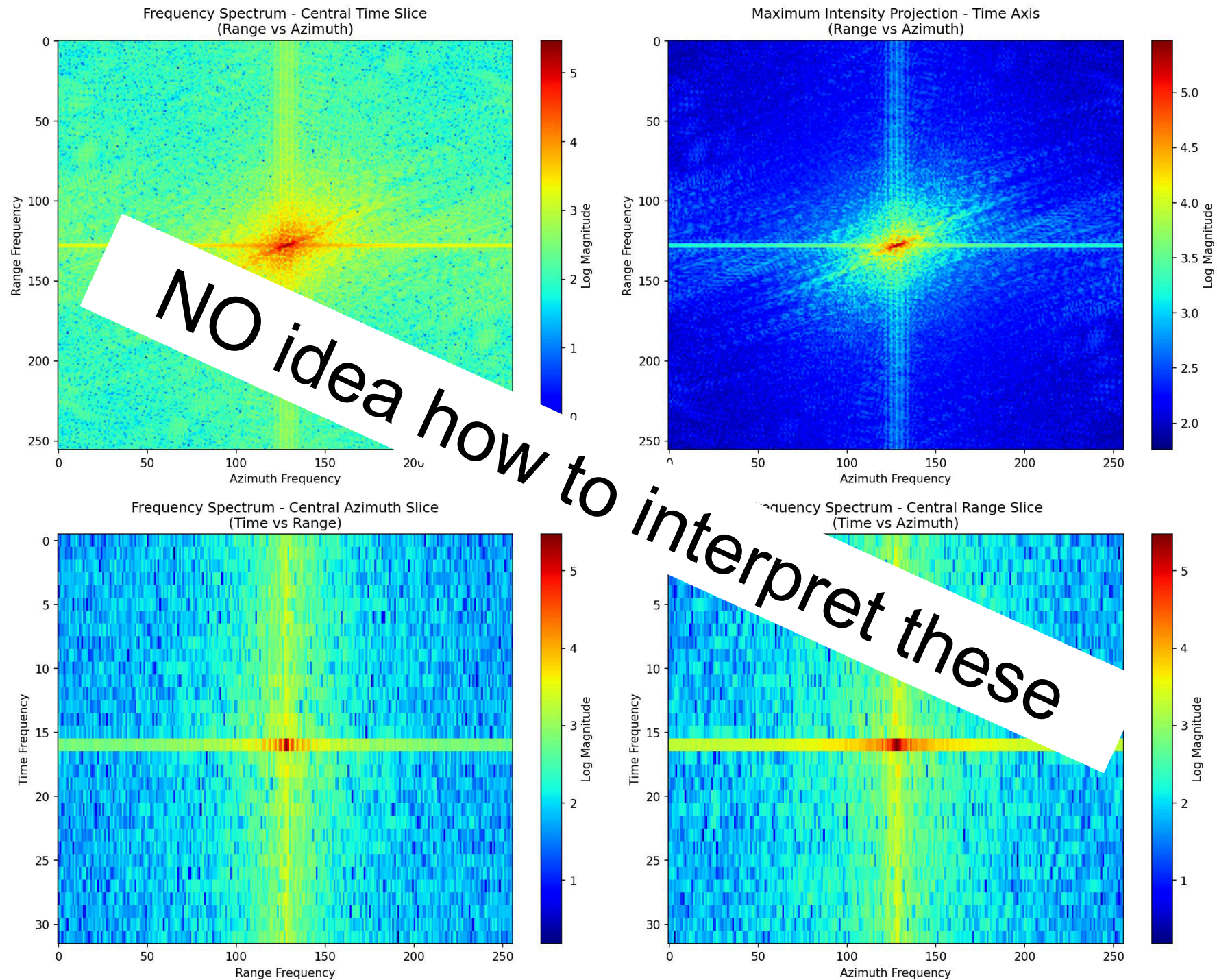
DETECTION ALGORITHMS

- Background subtraction with adaptive thresholding
- Frame-to-frame motion detection analysis
- Intensity-based radar reflector identification
- Hough transform for linear track detection

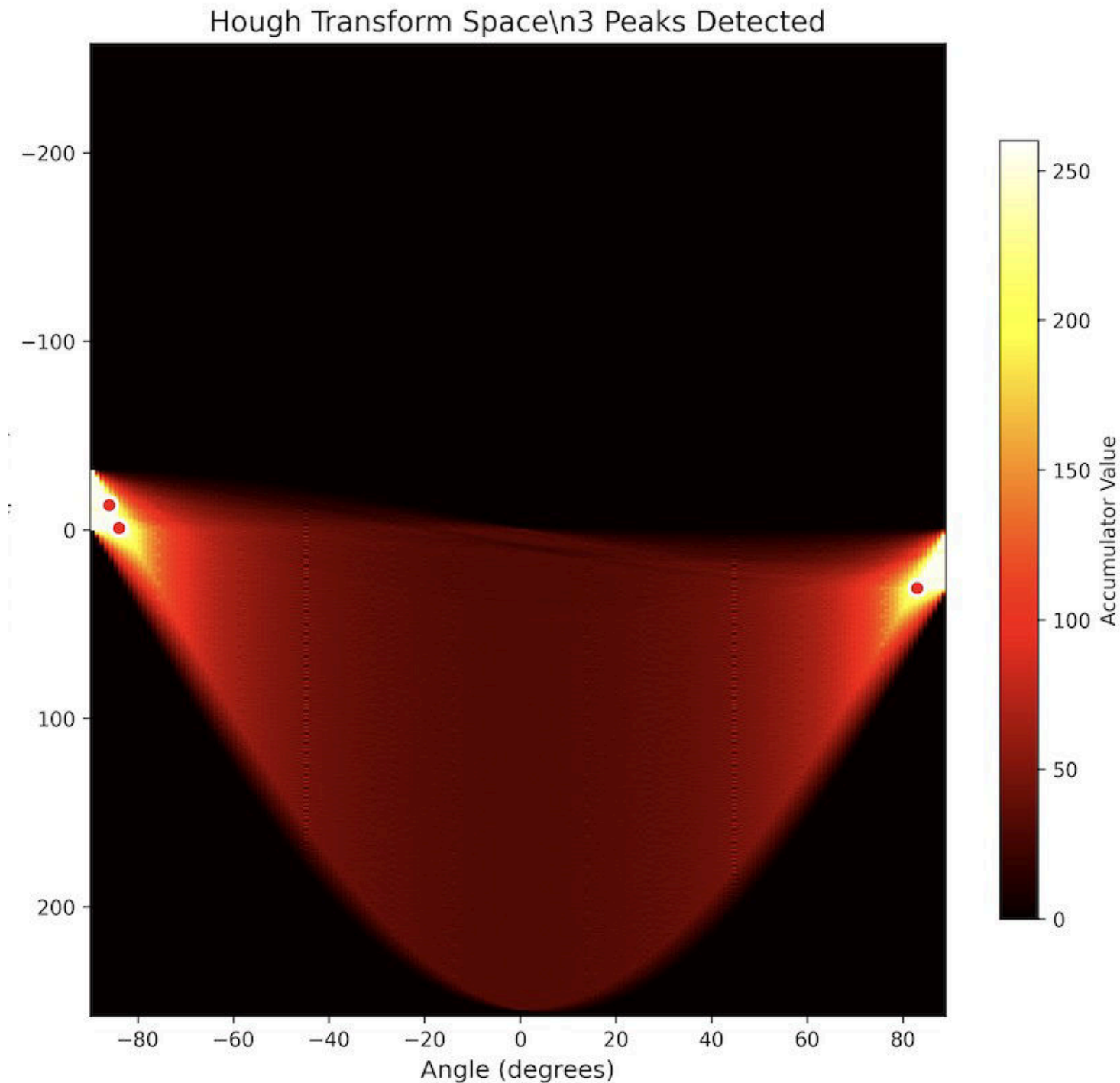
TRACKING & VISUALIZATION

- Multi-object tracking with bounding boxes
- Track persistence filtering and association
- Motion prediction and trajectory analysis

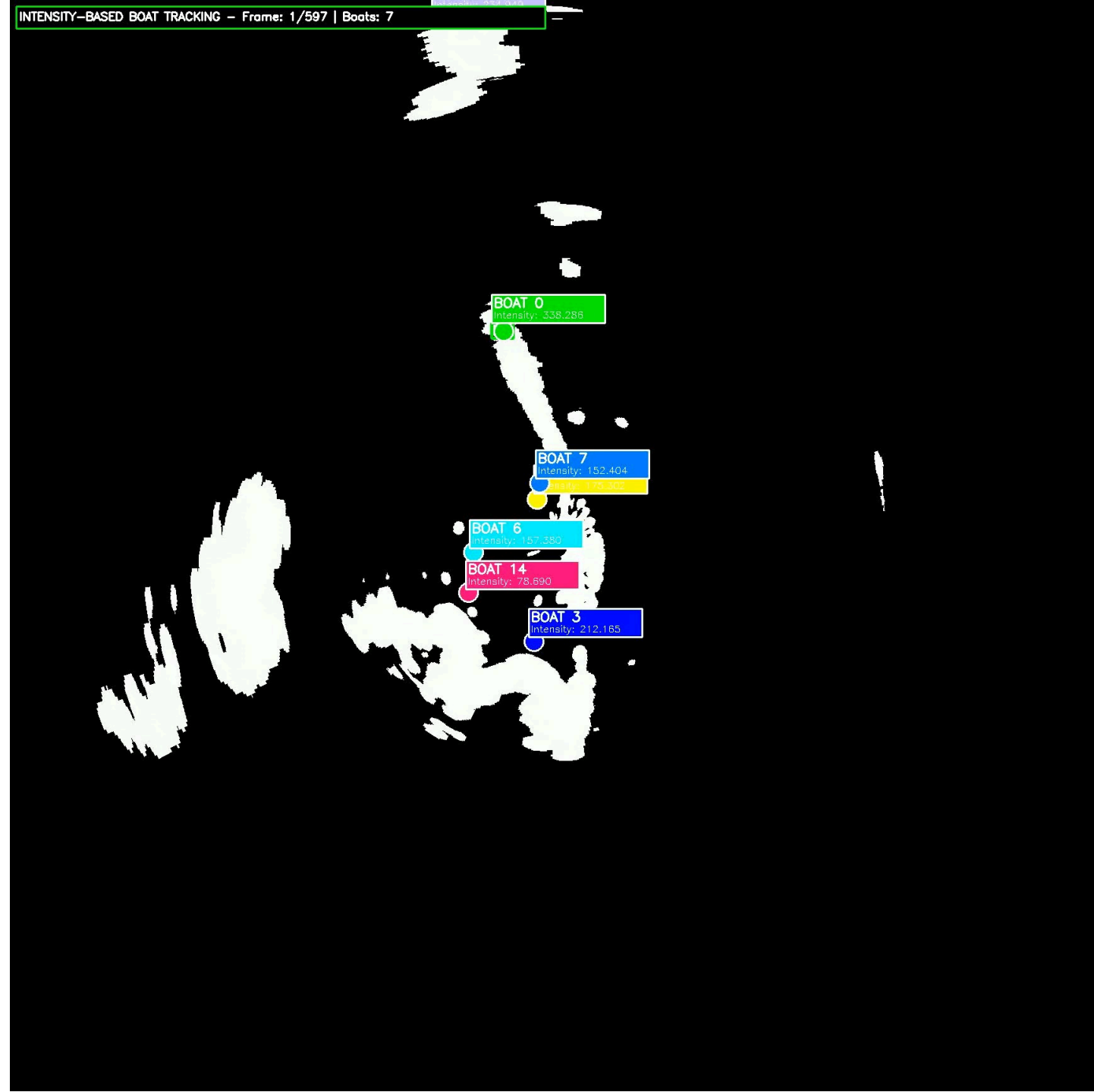
Object Tracking - FFT



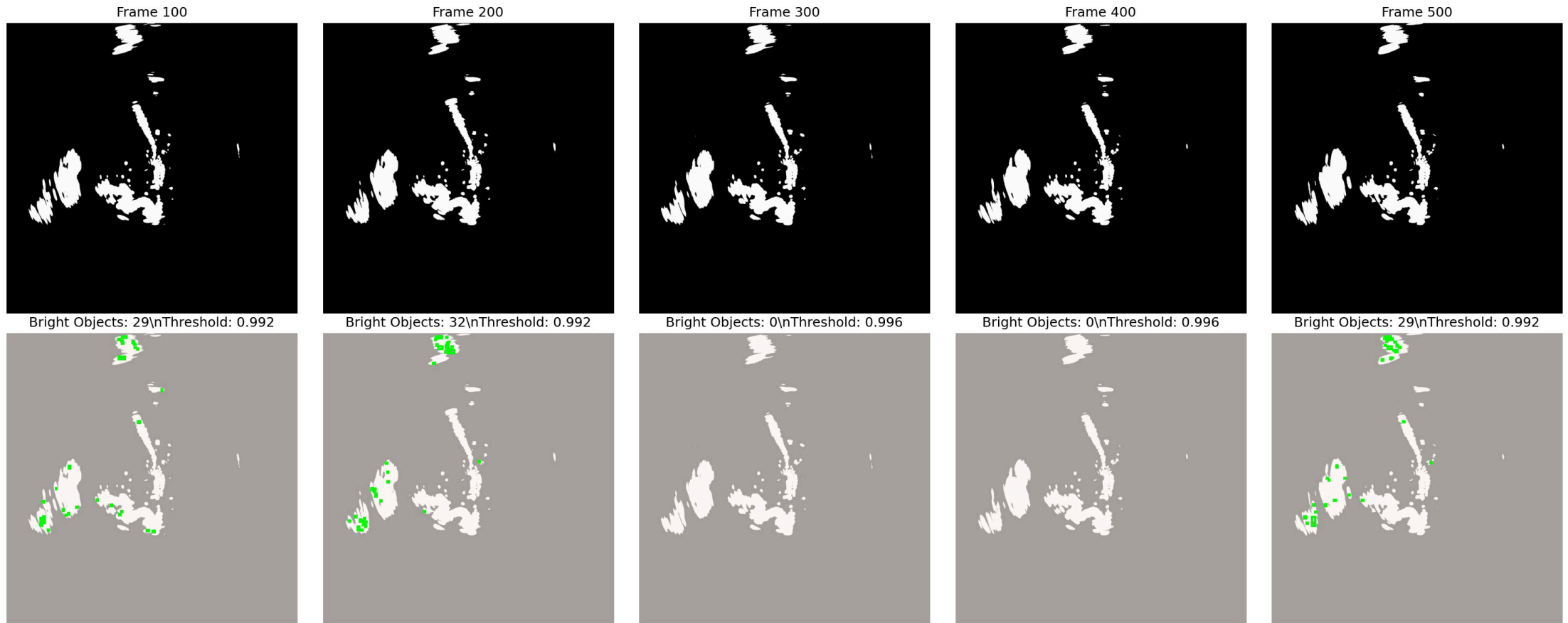
Object Tracking – Hough Transformation



Object Tracking – Intensity Based



Intensity

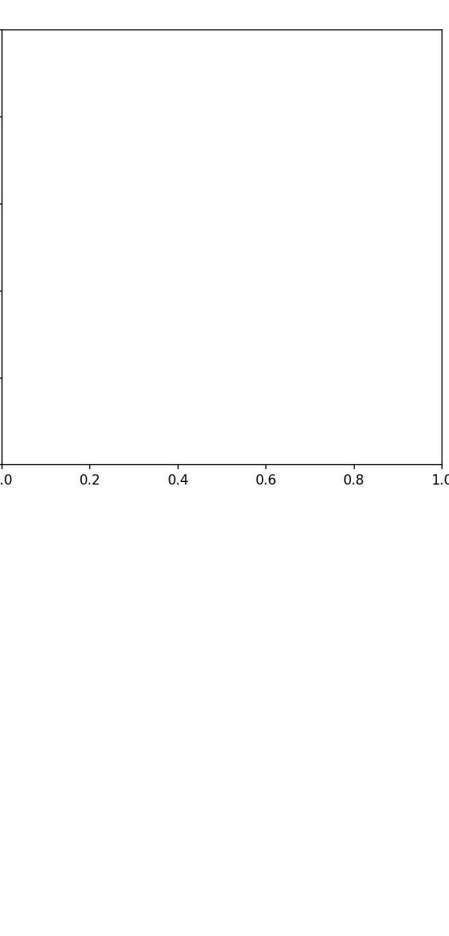
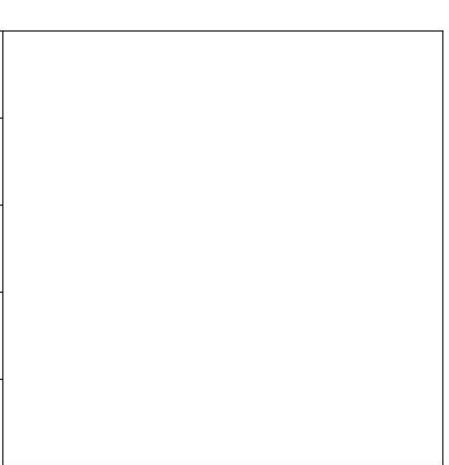
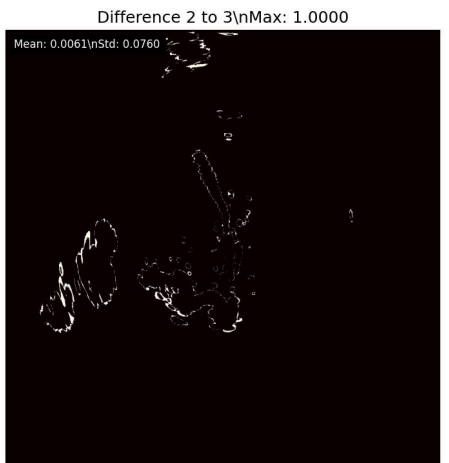
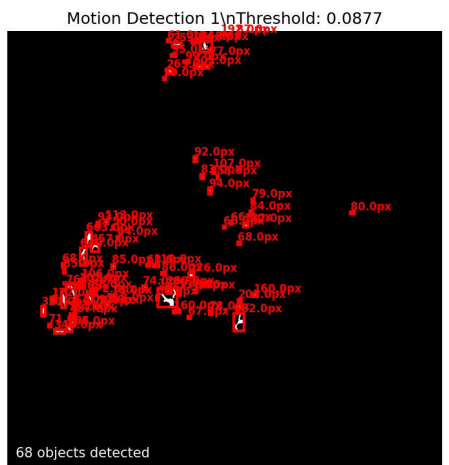
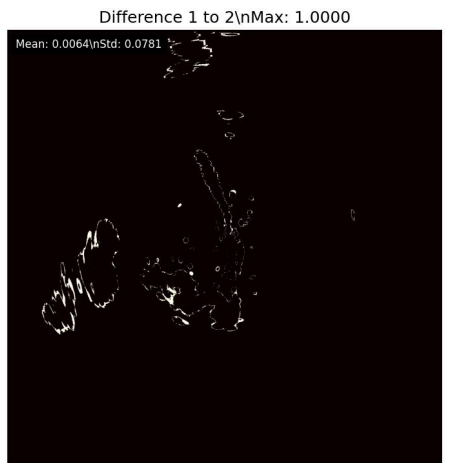
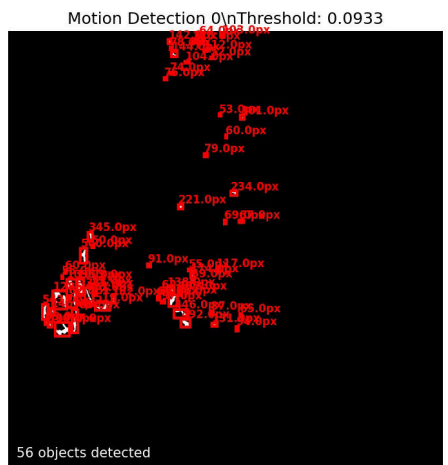
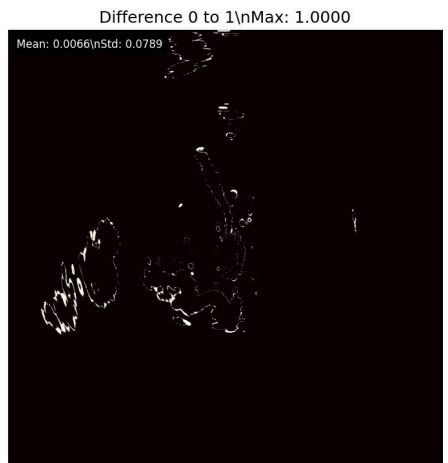


Object Tracking

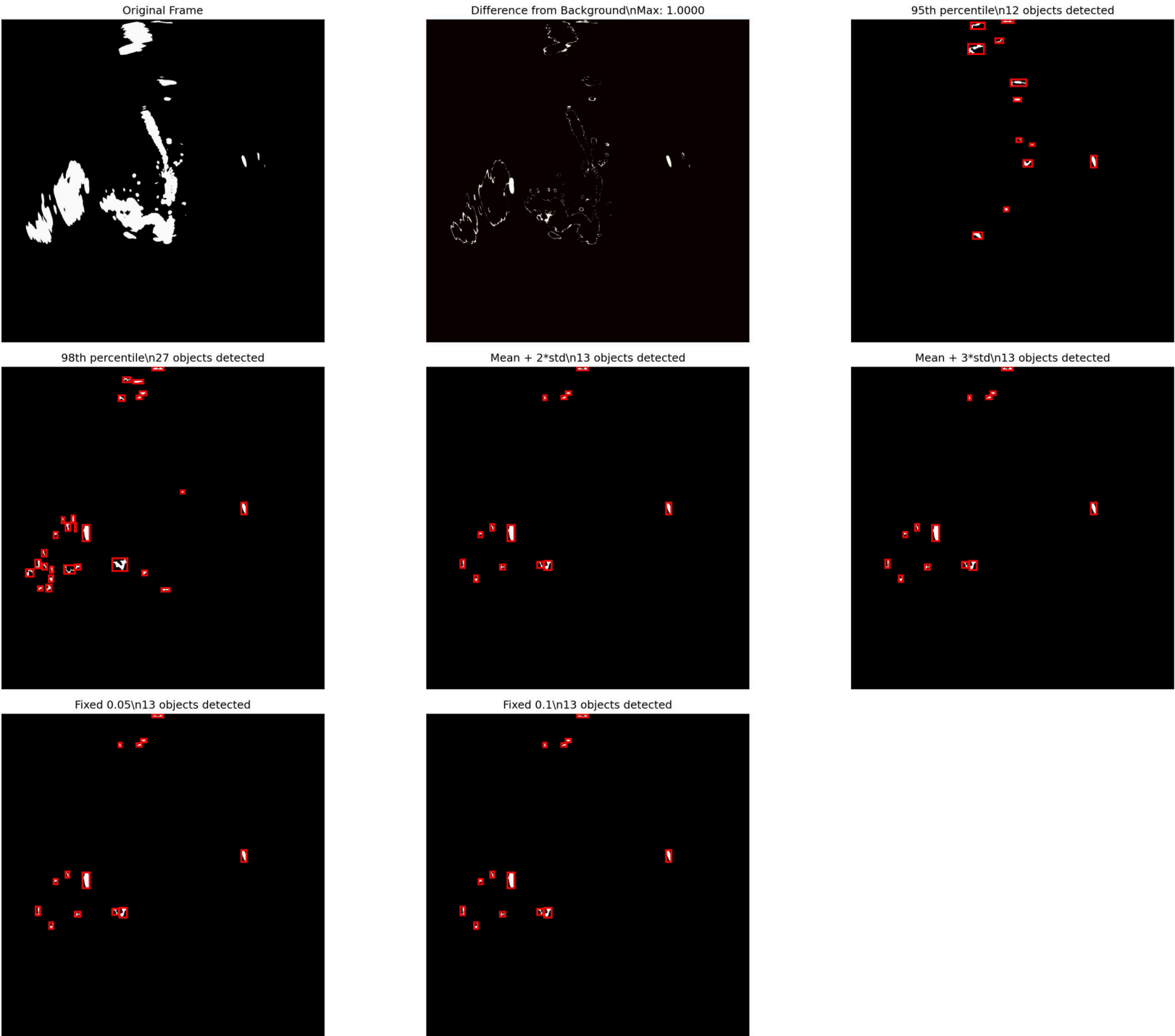
- Motion Based



Motion

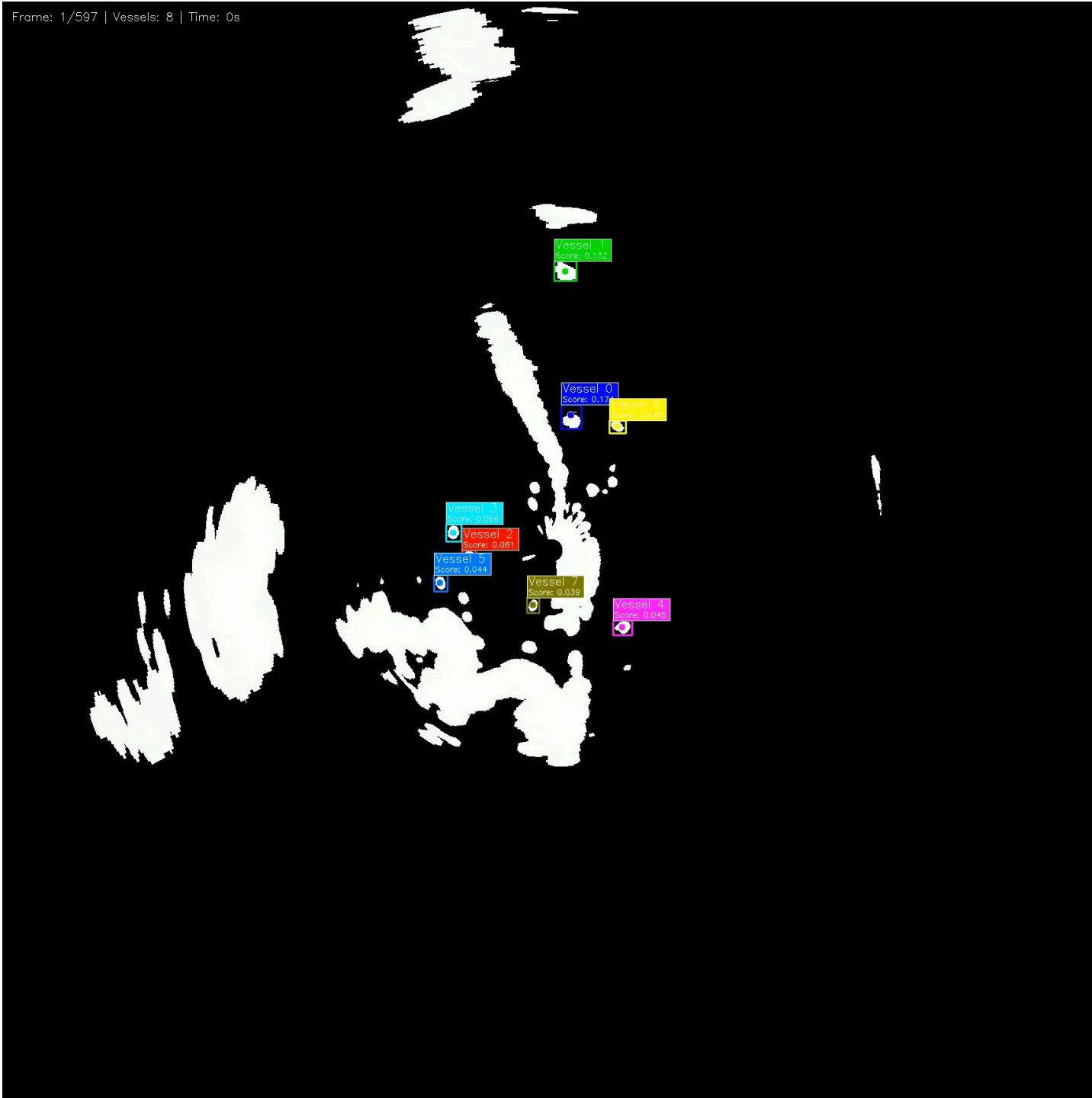


Motion



Object Tracking

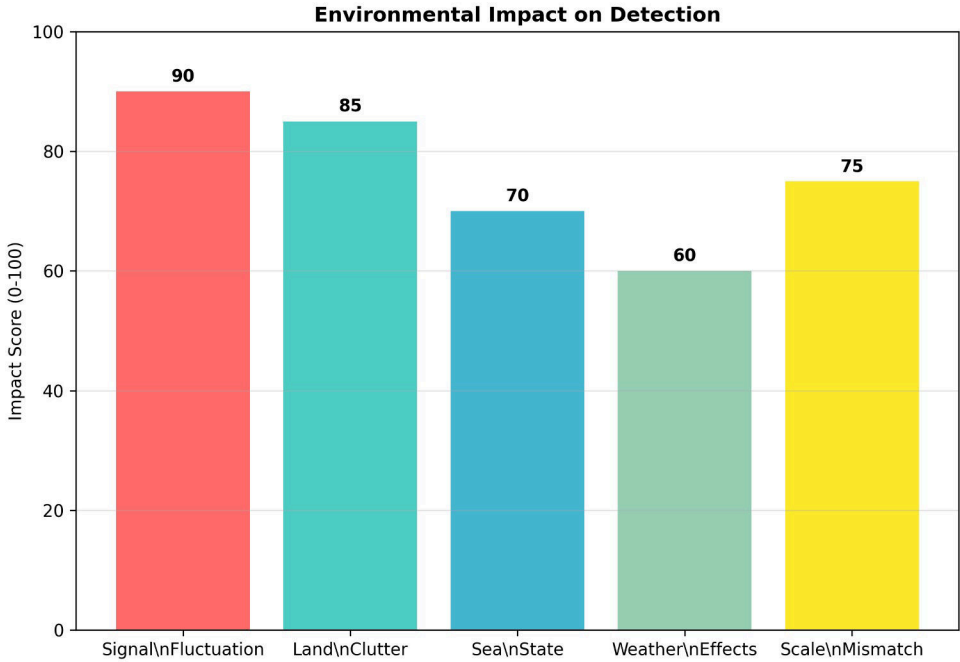
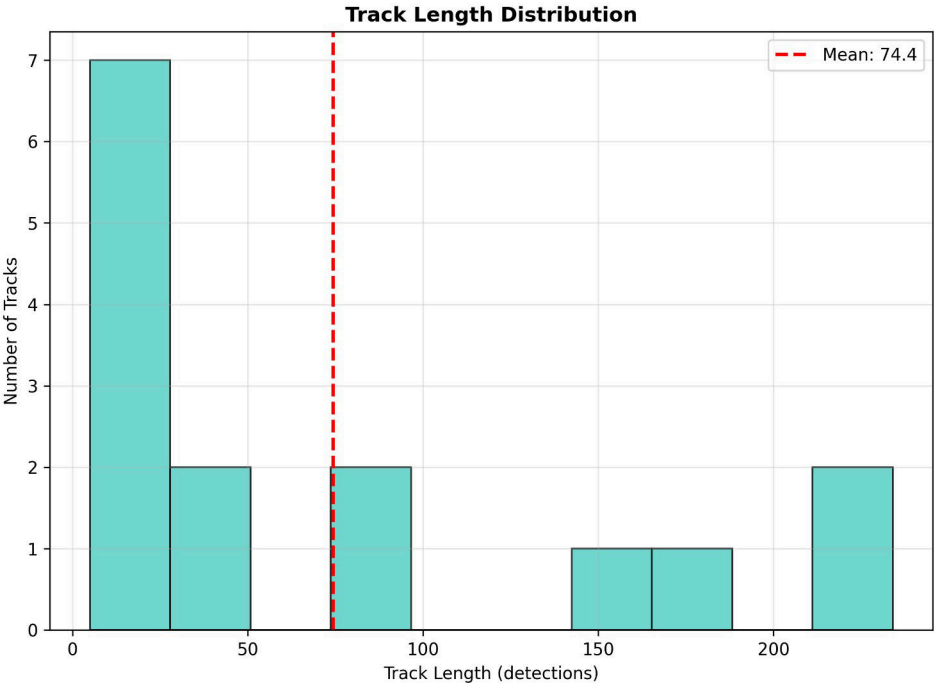
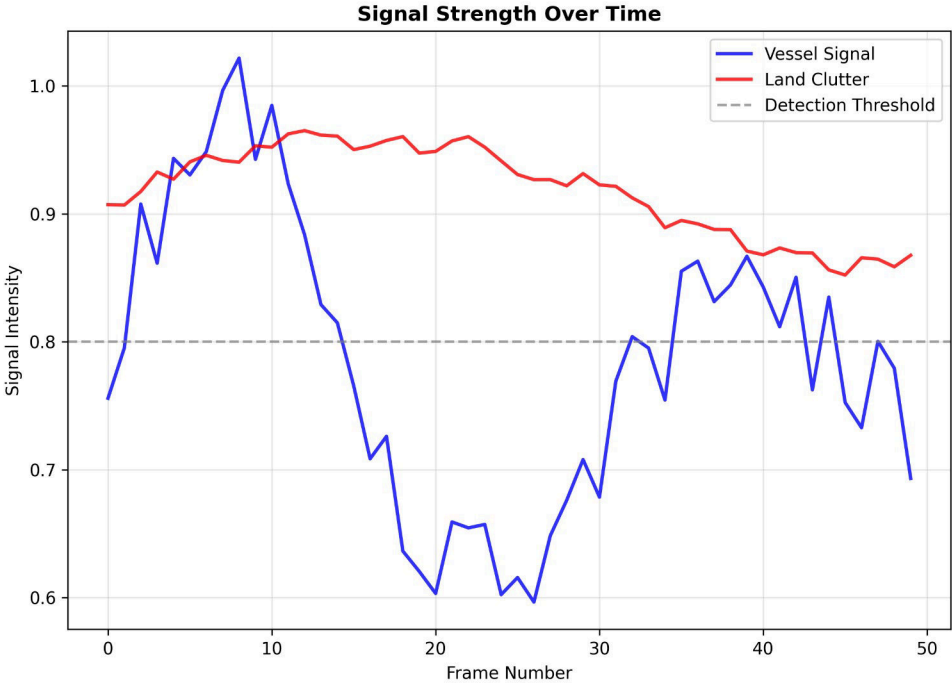
- Don't Remember



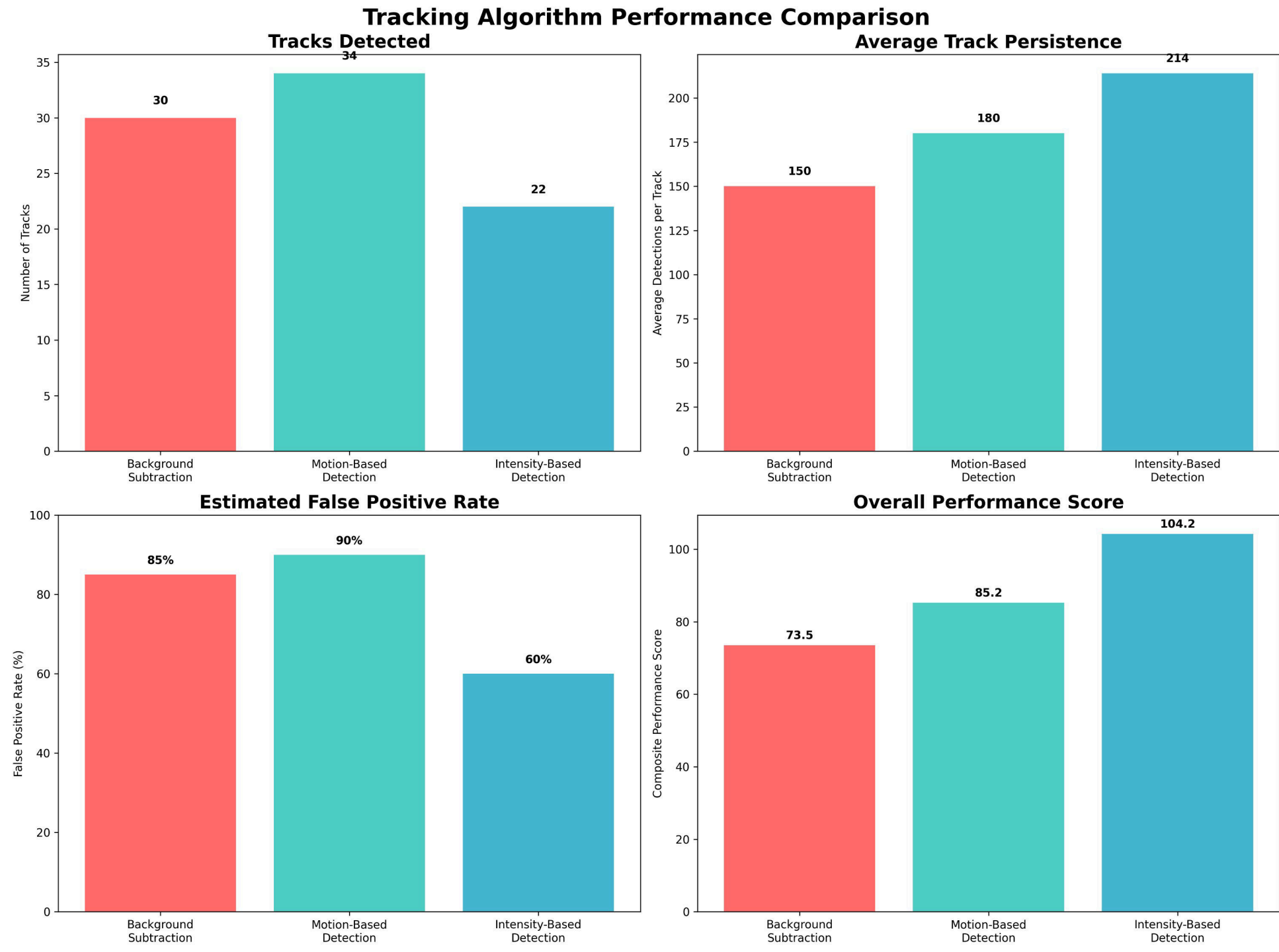
Object Tracking
– Don't
Remember

summary

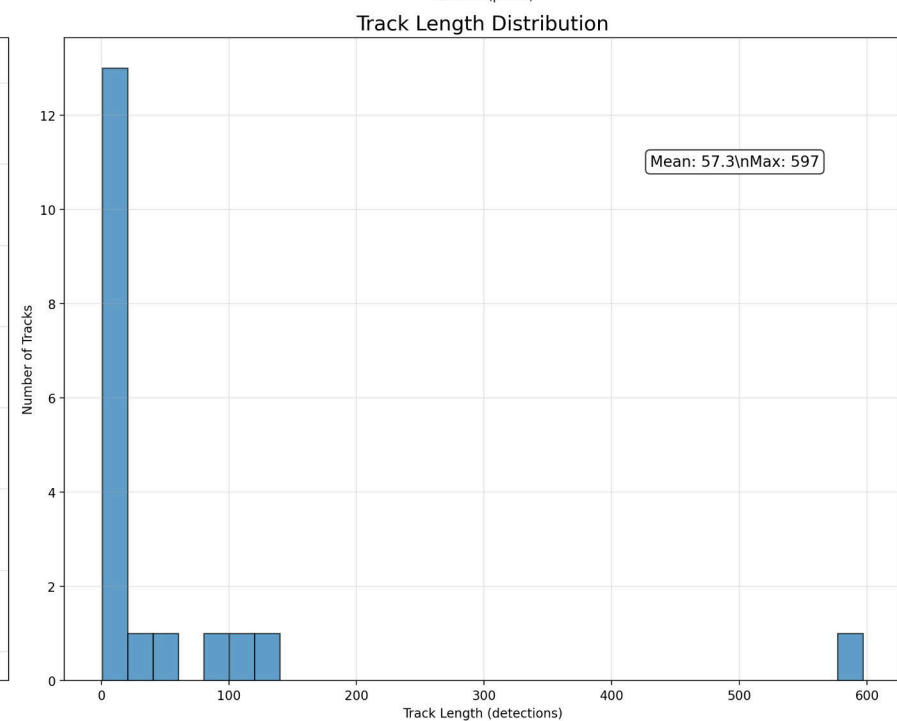
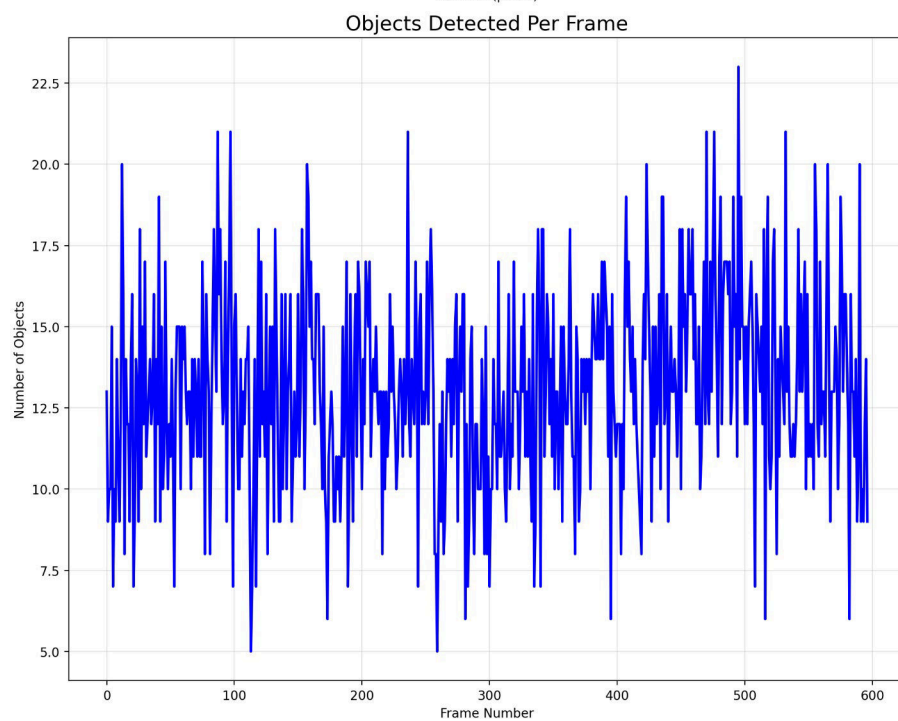
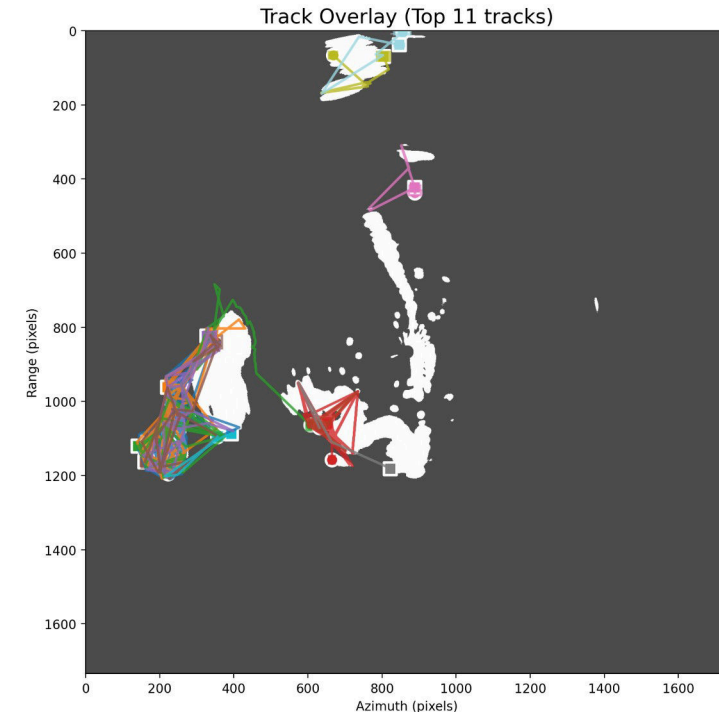
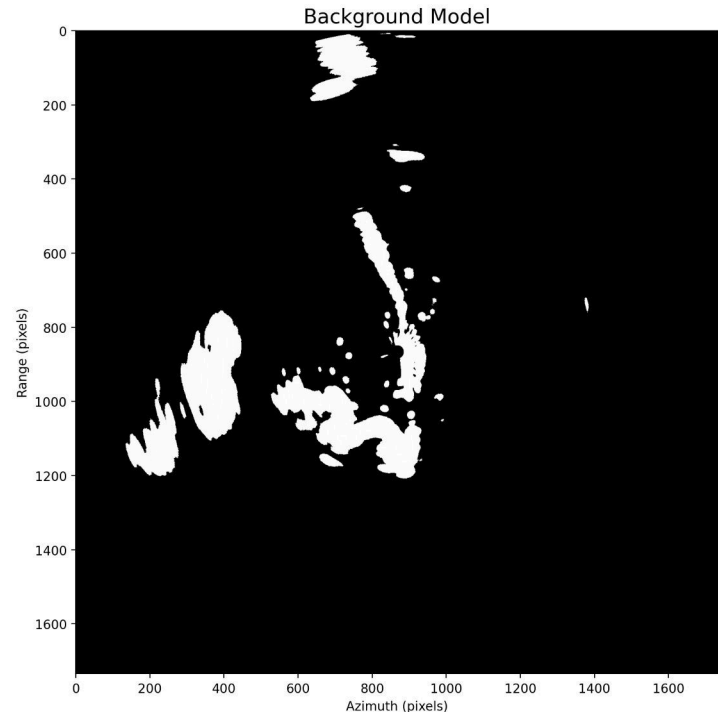
Radar Detection Challenges Analysis



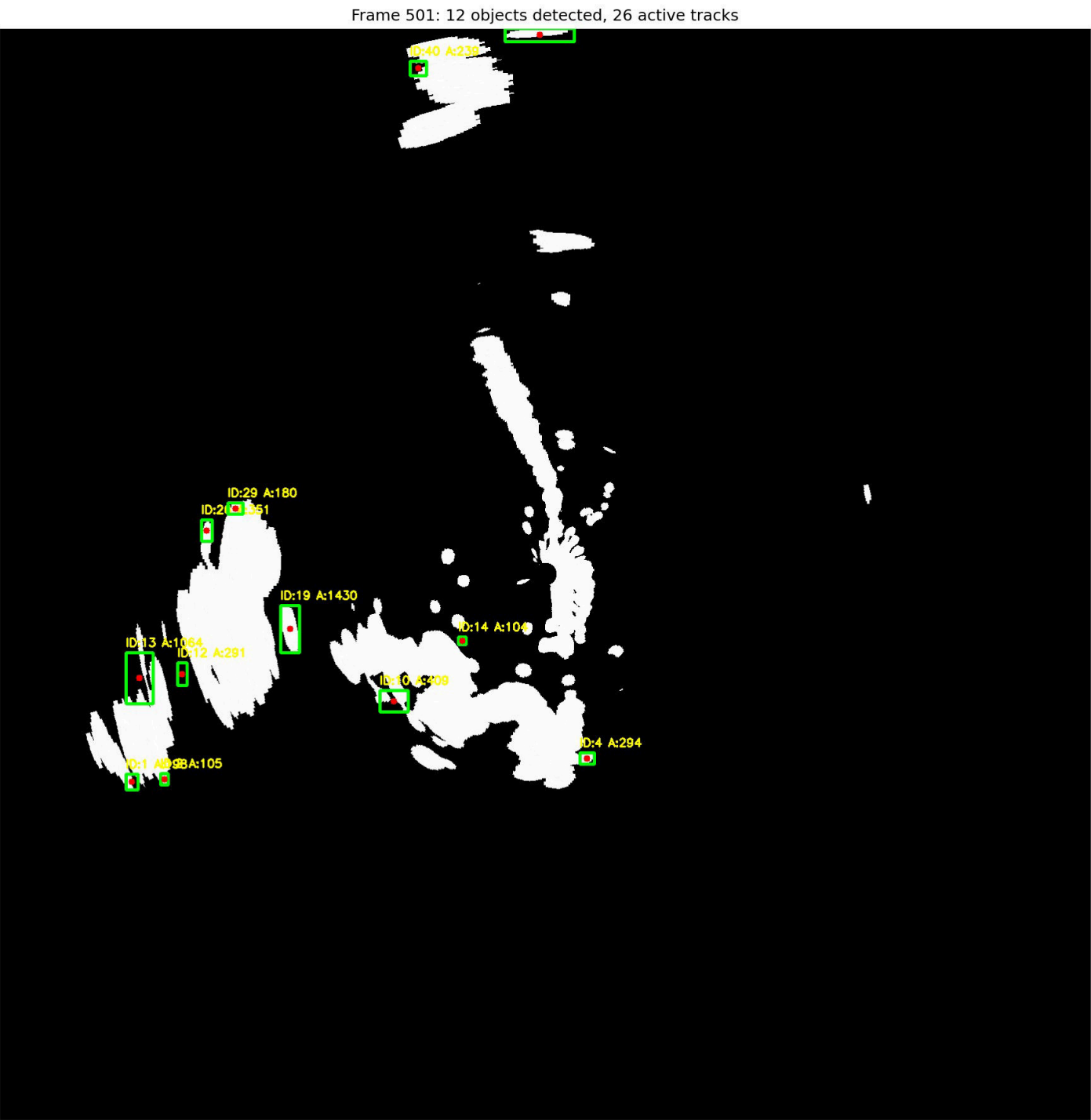
summary



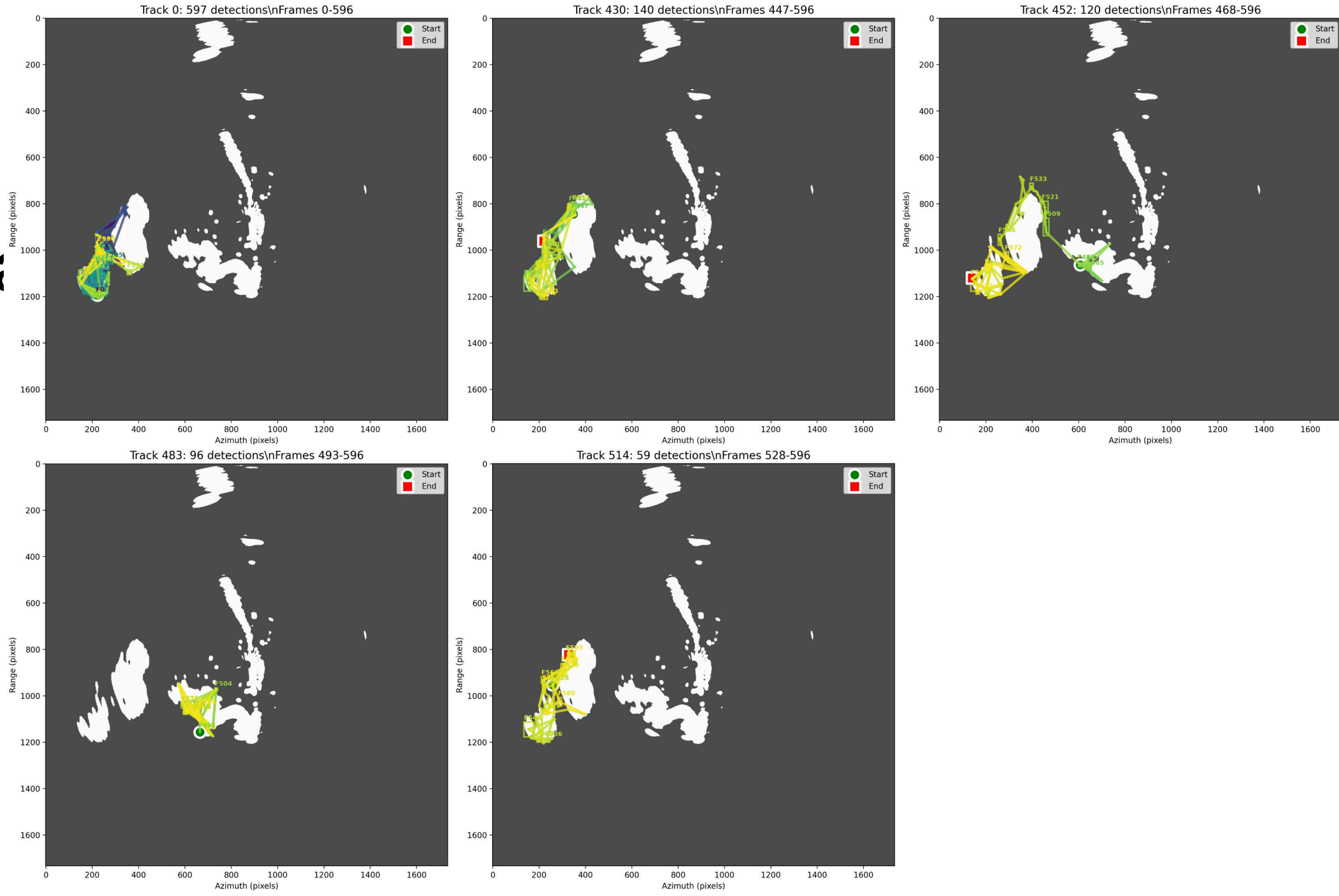
summary



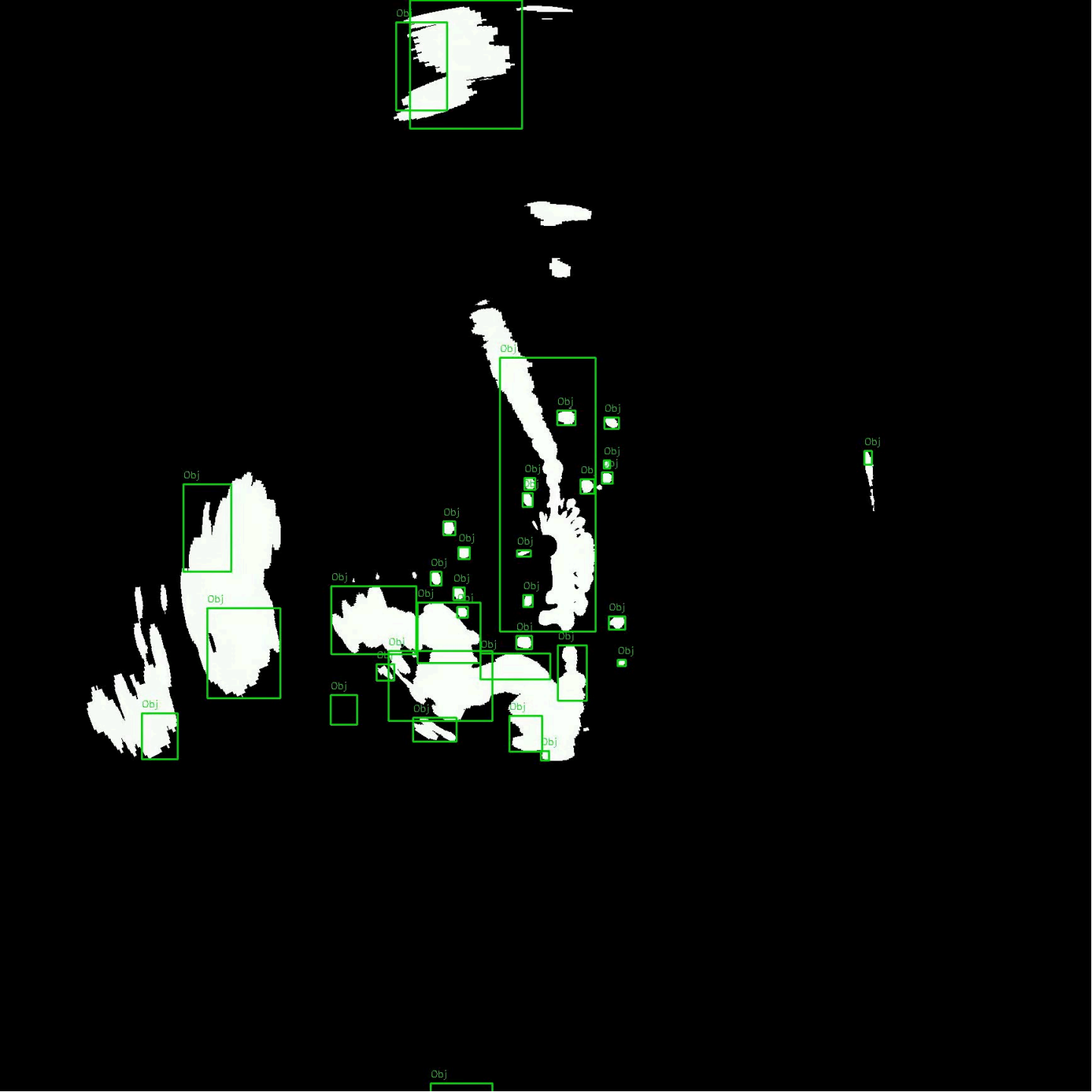
summary



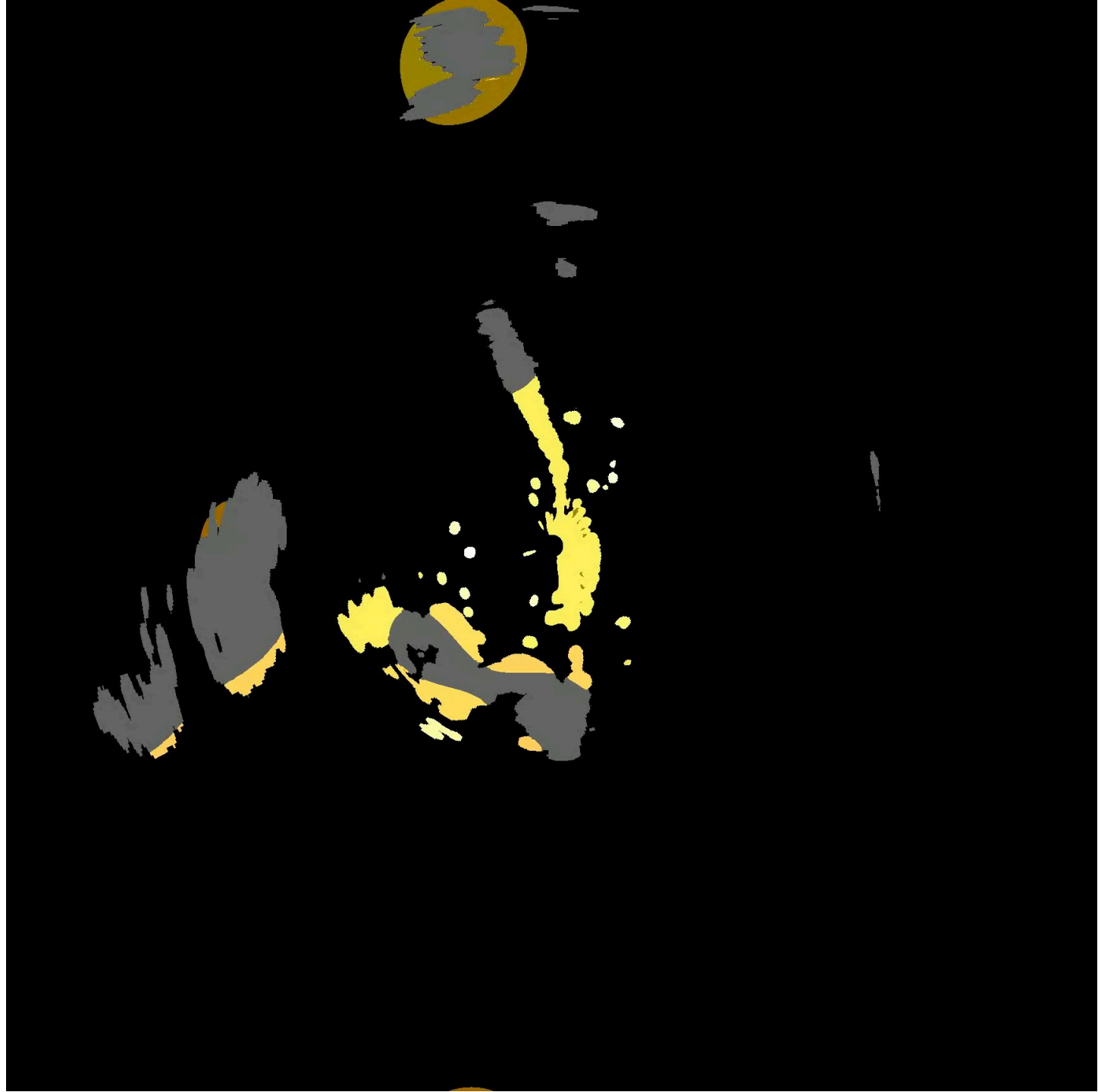
summary



summary



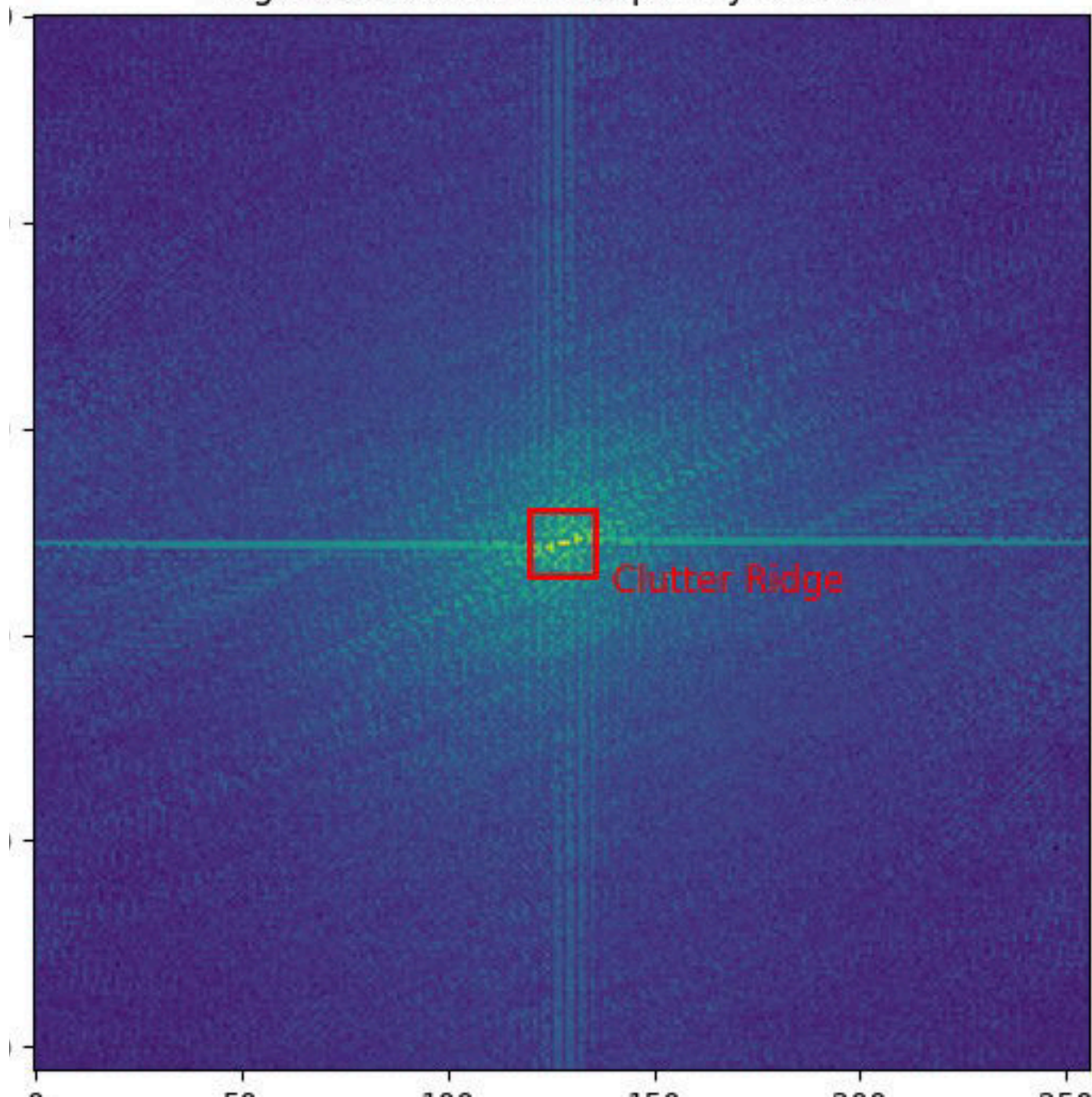
summary



summary

summary

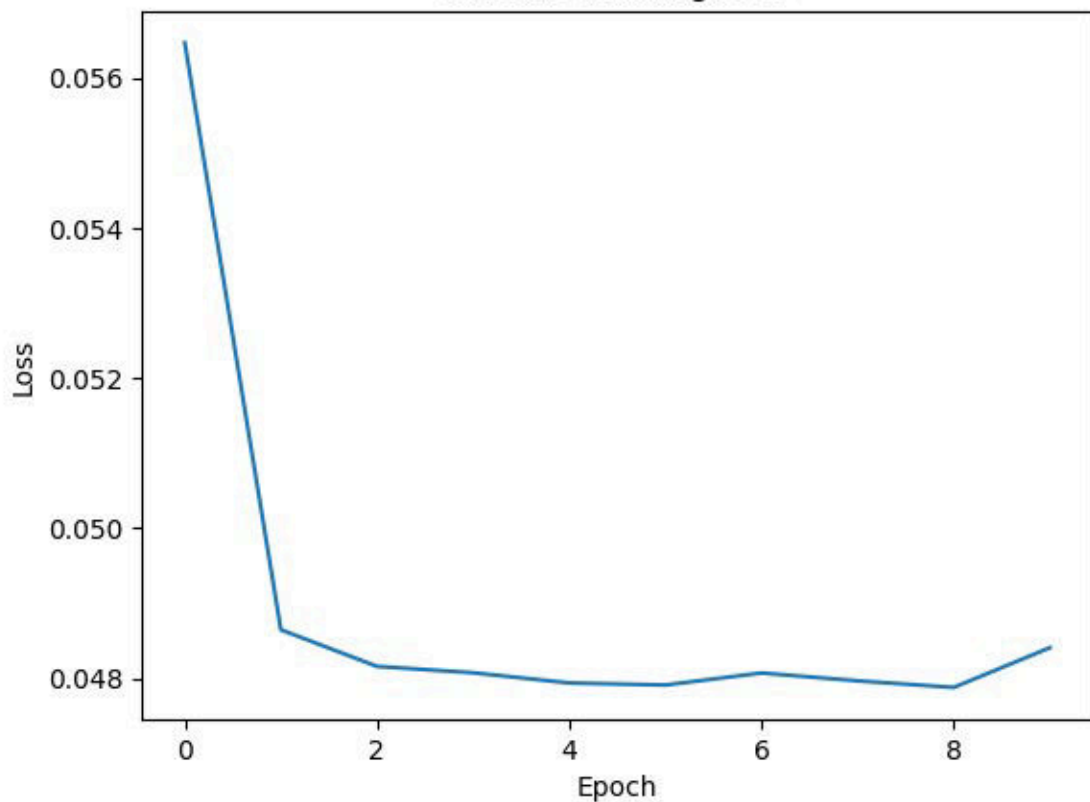
summary



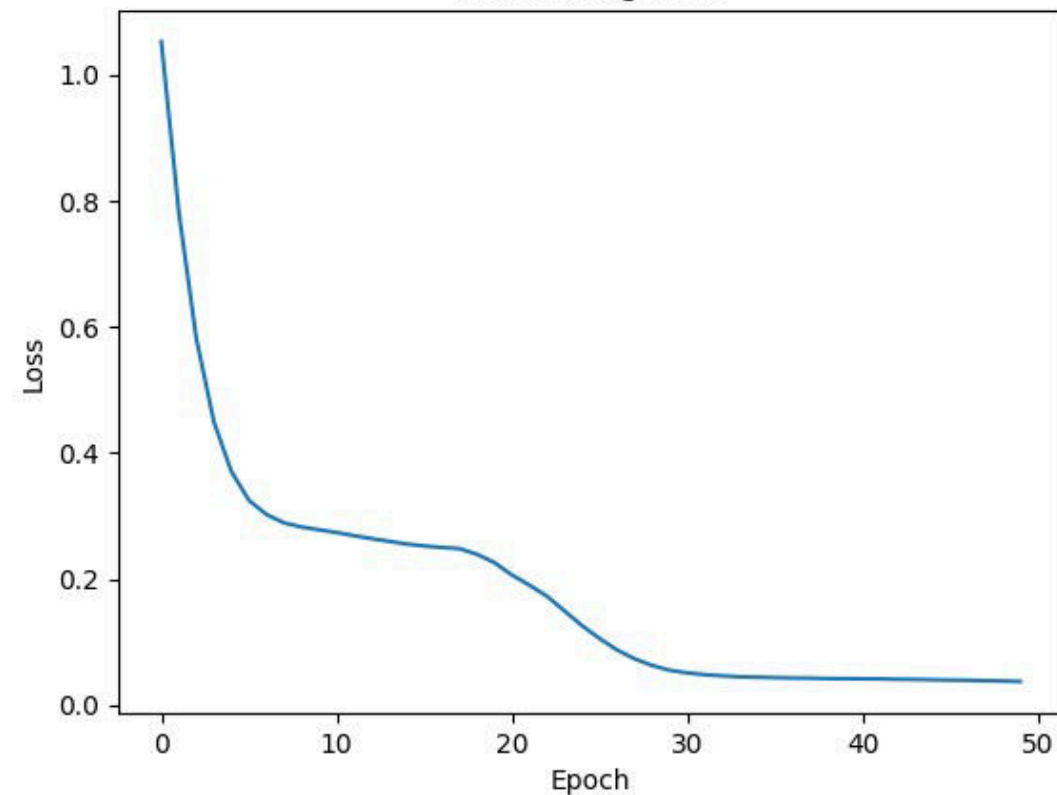
Tried building a temporal MAE at first using off-the-shelf designs.

MAE(feature extraction) → lightweight detection head(fine tuning)

MAE Pre-training Loss

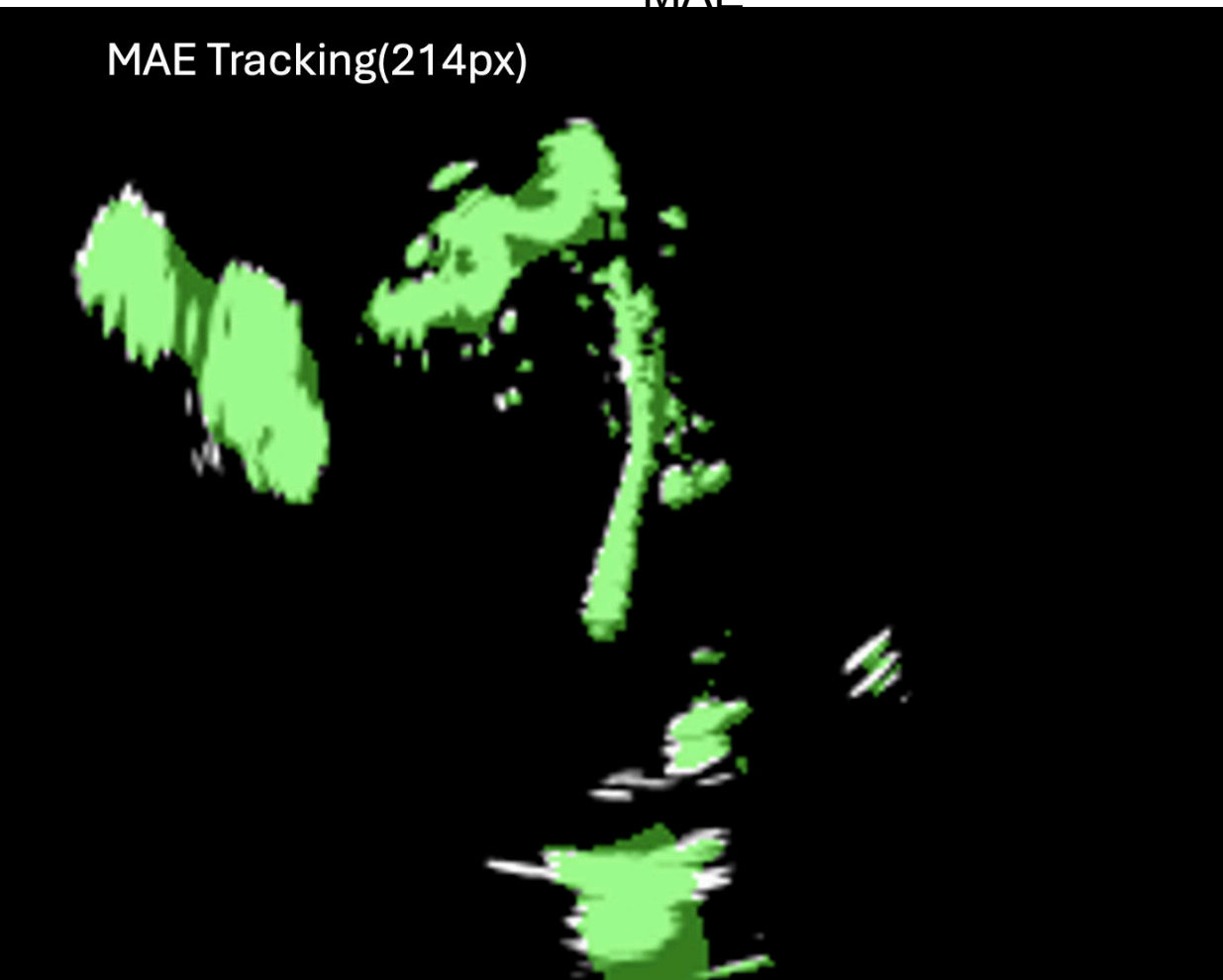


Fine-tuning Loss

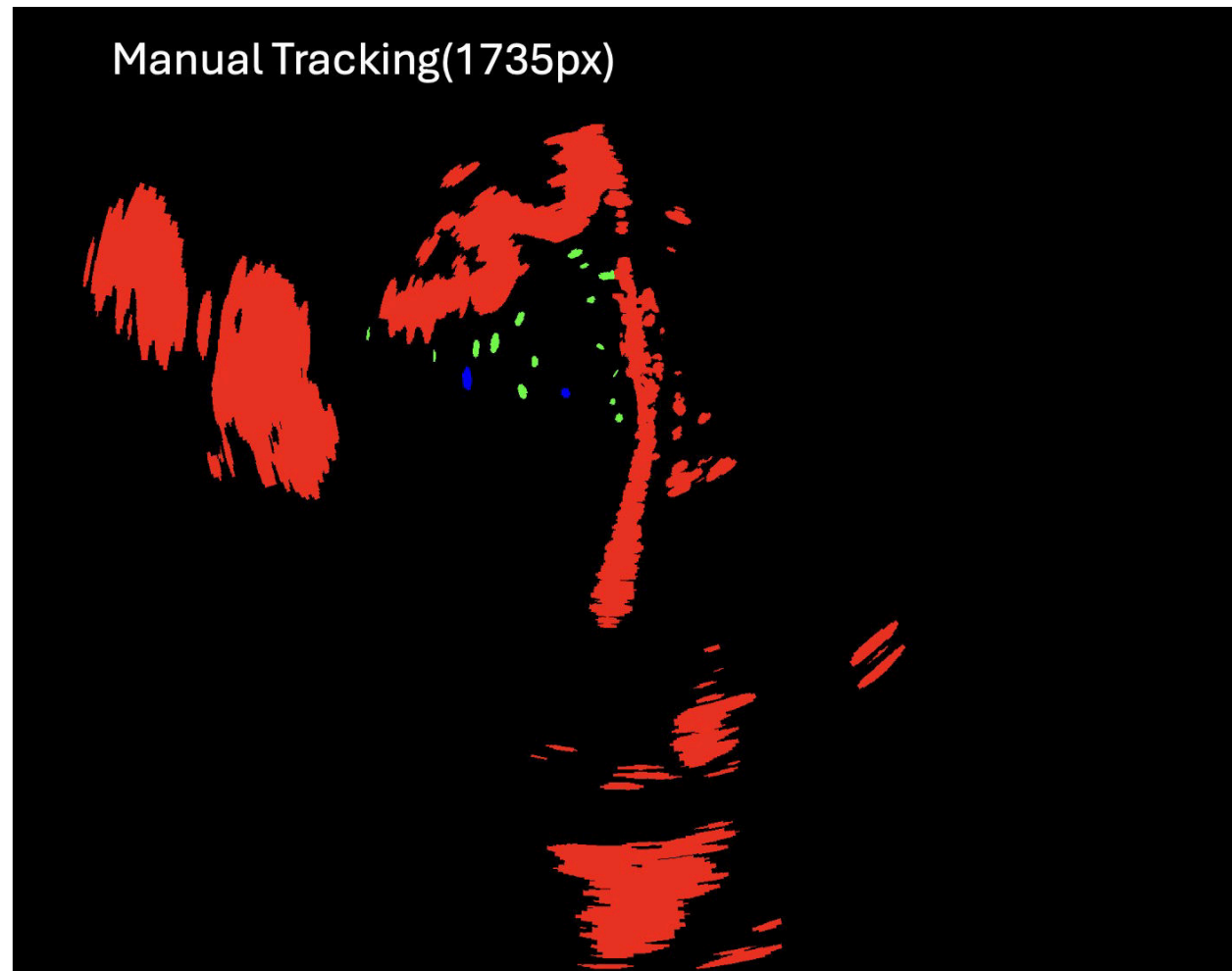


RESULTS FROM T- MAE

MAE Tracking(214px)



Manual Tracking(1735px)



MAE

Implementation

model_size	Encoder Embedding Dim	Encoder Depth	Encoder Heads	Decoder Embedding Dim	Decoder Depth	Decoder Heads
small	384	6	6	256	4	8
medium	512	8	8	384	6	12
large	768	12	12	512	8	16
xlarge	1024	16	16	768	12	24
xxlarge	1280	20	20	1024	16	32
huge	1536	24	24	1280	20	40
gigantic	2048	32	32	1536	24	48

MAE

Implementation

Parameter	Description	Default Value
<code>img_size</code>	The size of the input image (height and width).	32
<code>patch_size</code>	The size of each image patch.	4
<code>in_chans</code>	The number of input channels (e.g., 3 for RGB).	3
<code>embed_dim</code>	The embedding dimension of the encoder.	768
<code>depth</code>	The number of Transformer blocks in the encoder.	12
<code>num_heads</code>	The number of attention heads in the encoder.	12
<code>decoder_embed_dim</code>	The embedding dimension of the decoder.	512
<code>decoder_depth</code>	The number of Transformer blocks in the decoder.	8
<code>decoder_num_heads</code>	The number of attention heads in the decoder.	16
<code>mlp_ratio</code>	The ratio of the MLP hidden dimension to the embedding dimension.	4.0
<code>norm_layer</code>	The normalization layer to use.	<code>nn.LayerNorm</code>

MAE

Implementation

Parameter	Description	Default Value
num_epochs	The total number of training epochs.	50
learning_rate	The initial learning rate for the optimizer.	1e-4
mask_ratio	The ratio of patches to mask during training.	0.75
save_every	The frequency (in epochs) at which to save model checkpoints.	5
visualize_every	The frequency (in epochs) at which to generate and save reconstruction visualizations.	5

Parameter	Description	Default for adamw
weight_decay	The weight decay (L2 penalty).	0.05
betas	Coefficients used for computing running averages of gradient and its square.	(0.9, 0.999)
eps	Term added to the denominator to improve numerical stability.	1e-8

MAE Implementation

<code>scheduler_type</code>	Description
<code>cosine</code>	Cosine annealing scheduler (default).
<code>linear</code>	Linear decay scheduler.
<code>step</code>	Step decay scheduler.
<code>multistep</code>	Multi-step decay scheduler.
<code>exponential</code>	Exponential decay scheduler.
<code>plateau</code>	Reduce learning rate on plateau.
<code>warmup_cosine</code>	Cosine annealing with a warmup period.
<code>polynomial</code>	Polynomial decay scheduler.

<code>optimizer_type</code>	Description
<code>adamw</code>	AdamW optimizer (default).
<code>adam</code>	Adam optimizer.
<code>sgd</code>	Stochastic Gradient Descent (SGD) optimizer.
<code>rmsprop</code>	RMSprop optimizer.

MAE Implementation

MAE ARCHITECTURE DEEP DIVE

ENCODER (ASYMMETRIC)

- **Input:** Only visible patches (~25%)
- **Architecture:** Vision Transformer
- **Purpose:** Extract features from visible content
- **Efficiency:** Processes only 25% of tokens

```
visible_patches =  
select_visible(patches) encoded =  
transformer_encoder(visible_patches)
```

DECODER (LIGHTWEIGHT)

- **Input:** Encoded patches + mask tokens
- **Architecture:** Smaller Transformer
- **Purpose:** Reconstruct masked patches
- **Output:** Pixel predictions for masked areas

```
full_sequence = encoded +  
mask_tokens reconstructed =  
transformer_decoder(full_sequence)
```

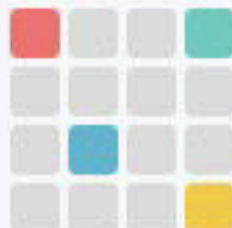
WHAT IS A MASKED AUTOENCODER?

MAE CORE CONCEPT

INPUT IMAGE



MASKED (75% HIDDEN)



RECONSTRUCTION



Self-supervised learning: Learn visual representations by reconstructing randomly masked image patches

MAE Implementation

MODEL ARCHITECTURE CONFIGURATIONS

SMALL MODEL

14M

Parameters

6 Encoder Layers

4 Decoder Layers

384 Embed Dim

MEDIUM MODEL

36M

Parameters

8 Encoder Layers

6 Decoder Layers

512 Embed Dim

LARGE MODEL

111M

Parameters

12 Encoder Layers

8 Decoder Layers

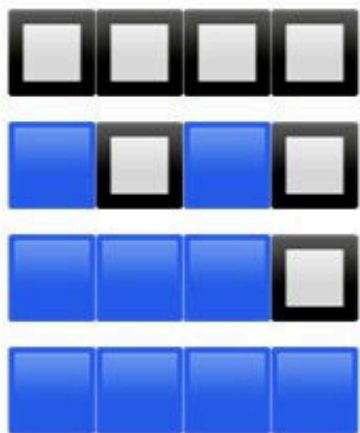
768 Embed Dim

Each model tested with masking ratios: 35%, 50%, 60%, 75%, 90%

MAE Implementation

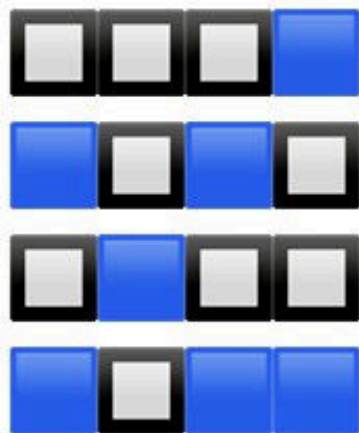
MASKING STRATEGY IMPACT

35% MASKED



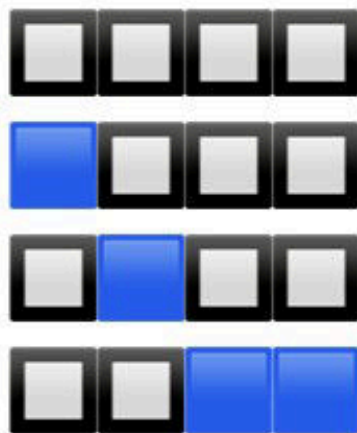
Easier reconstruction

50% MASKED



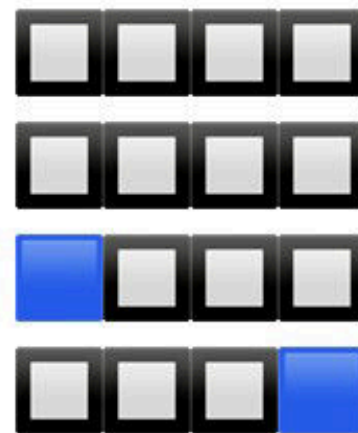
Balanced challenge

60% MASKED



More challenging

75% MASKED



Standard MAE ratio

90% MASKED



Extreme challenge

■ = Masked patches (model must predict) | ■ = Visible patches (encoder input)

EXPERIMENT MATRIX OVERVIEW

LARGE - 35%

0.189

Test Loss

1.1h | 110.8M params

LARGE - 50%

0.239

Test Loss

1.0h | 110.8M params

LARGE - 60%

0.296

Test Loss

0.9h | 110.8M params

LARGE - 75%

0.414

Test Loss

0.7h | 110.8M params

LARGE - 90%

0.709

Test Loss

0.6h | 110.8M params

MEDIUM - 35%

0.193

Test Loss

0.4h | 36.2M params

MEDIUM - 50%

0.244

Test Loss

0.4h | 36.2M params

MEDIUM - 60%

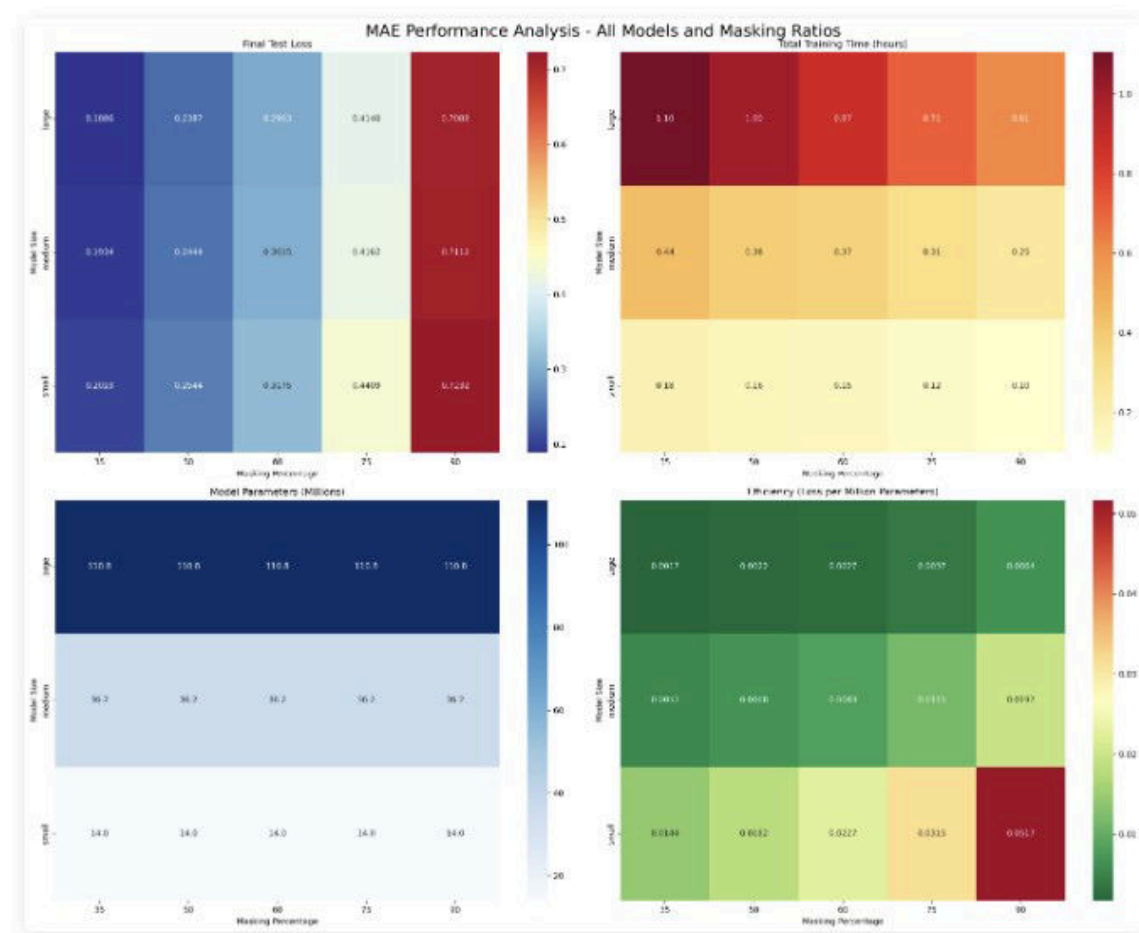
0.301

Test Loss

0.4h | 36.2M params

Complete systematic exploration: 15 experiments covering all size/masking combinations

BEST PERFORMING CONFIGURATIONS



BEST OVERALL

Large @ 35%

0.189 test loss



FASTEST TRAINING

Small @ 90%

5.9 hours

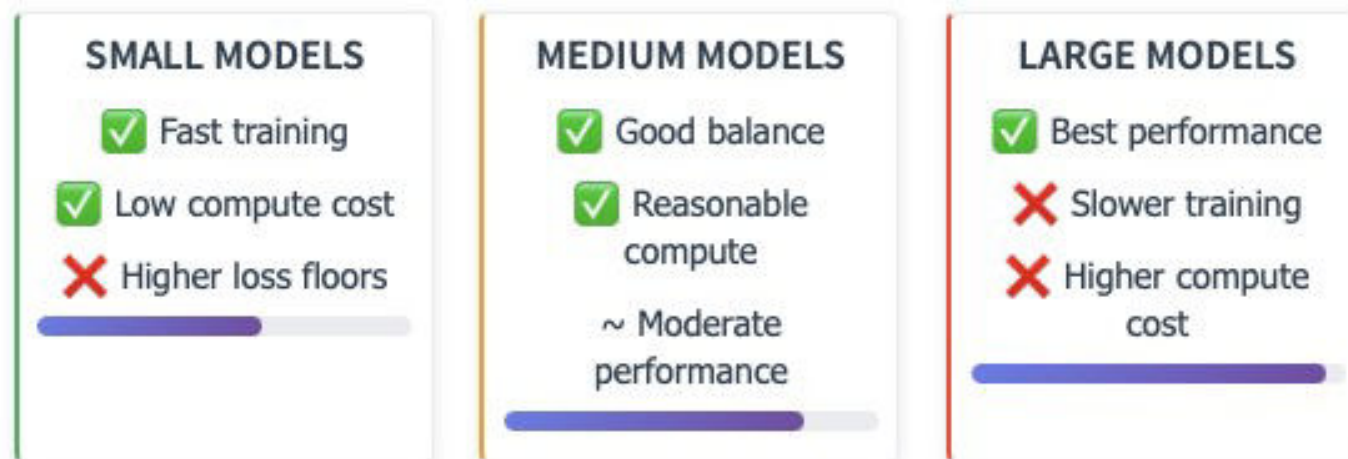


MOST EFFICIENT

Large @ 35%

Best loss/param ratio

MAE Implementation



KEY FINDING #2: MODEL SCALING BENEFITS



Diminishing returns observed: larger models help, but gains decrease with size

MAE Implementation

SMALL MODEL @ 35% MASKING

CONFIGURATION

Encoder: 6 layers

Decoder: 4 layers

Embed Dim: 384

Parameters: 14.0M

RESULTS

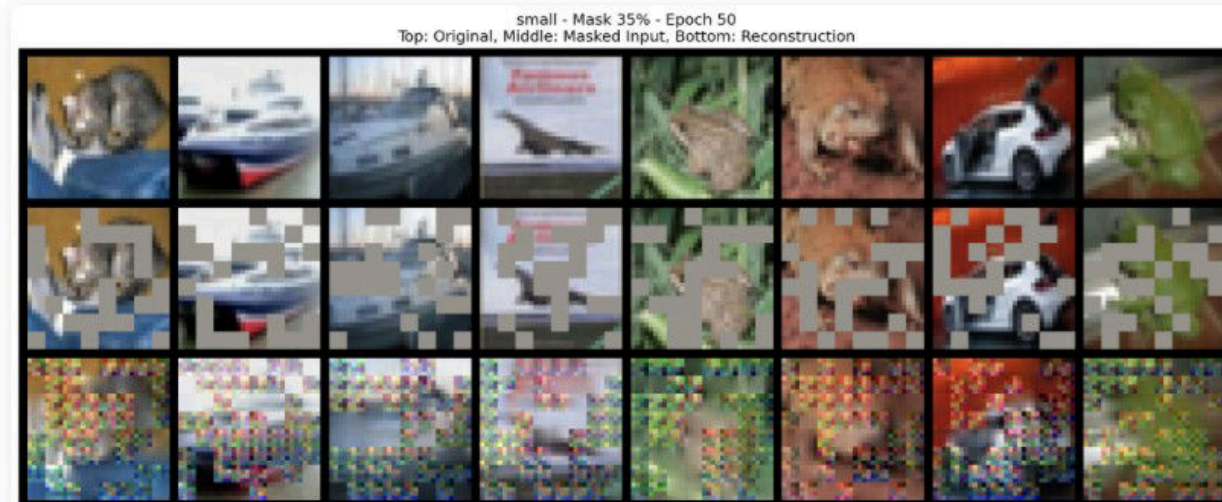
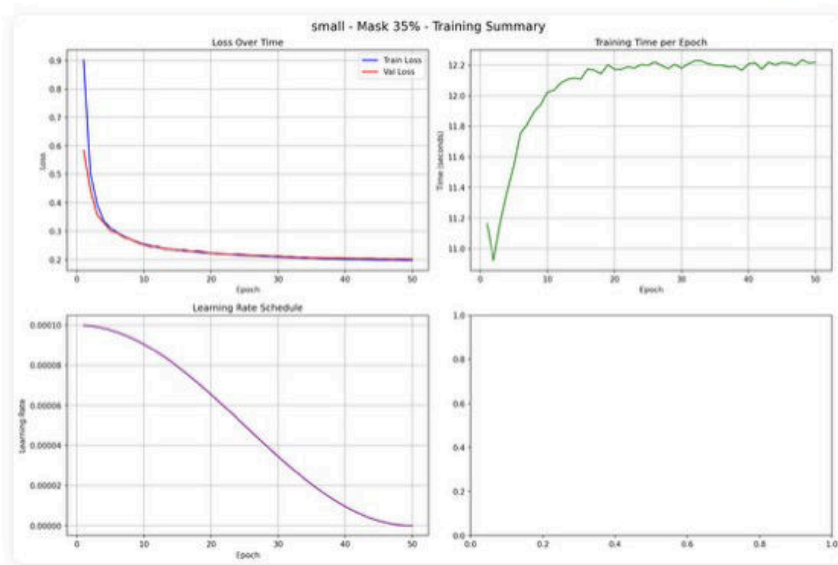
Test Loss: 0.202

Best Val Loss: 0.202

Training Time: 0.2h

Avg Epoch Time: 13.3s

TRAINING PROGRESS & FINAL RESULTS



MAE Implementation

SMALL MODEL @ 75% MASKING

CONFIGURATION

Encoder: 6 layers

Decoder: 4 layers

Embed Dim: 384

Parameters: 14.0M

RESULTS

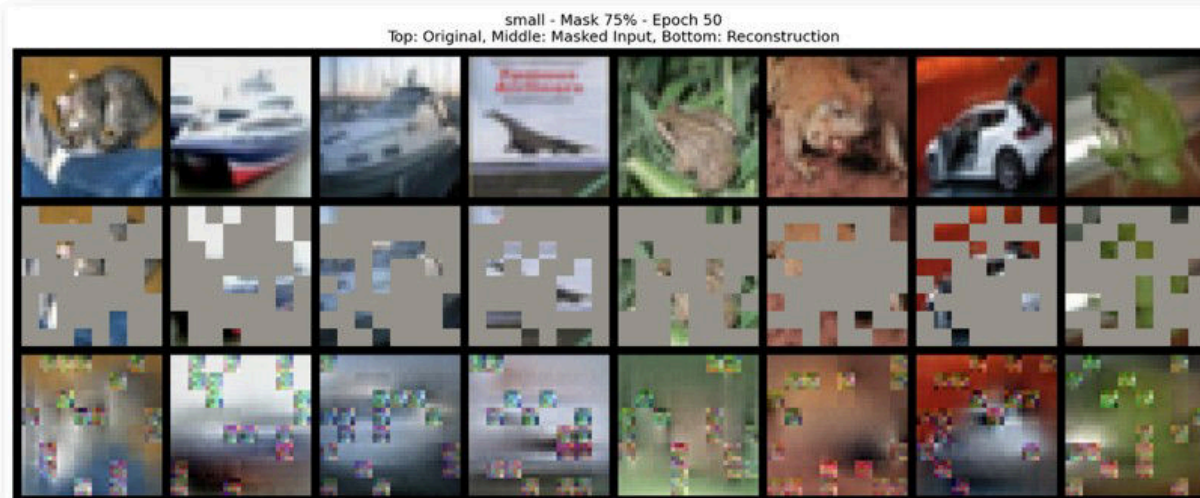
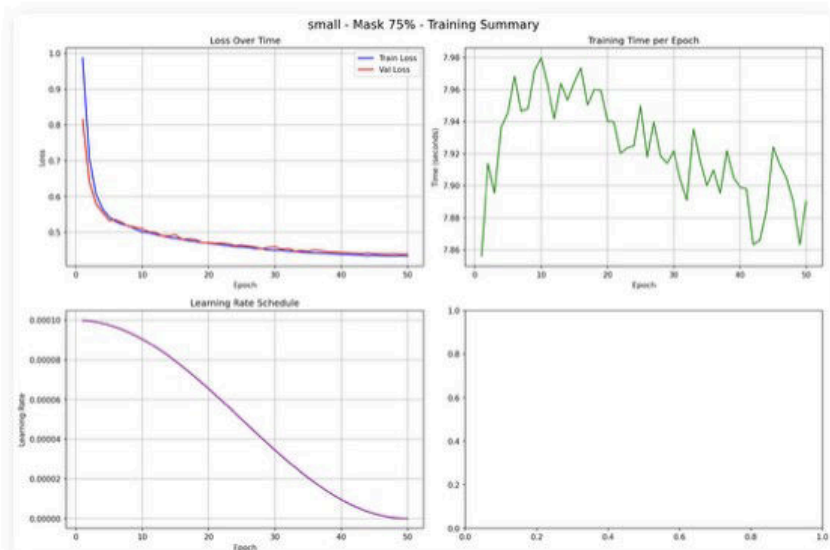
Test Loss: 0.441

Best Val Loss: 0.439

Training Time: 0.1h

Avg Epoch Time: 8.8s

TRAINING PROGRESS & FINAL RESULTS



MAE Implementation

GIGANTIC MAE MODEL - 2.3 BILLION PARAMETERS

GIGANTIC MODEL

2.3B

Parameters

32 Encoder Layers

24 Decoder Layers

2048 Embed Dim

32 Attention Heads

✓ Adaptive Learning Rate Scheduling

✓ Early Stopping with Loss Plateau Detection

✓ Hybrid Reconstruction Visualization

✓ Neural Network Introspection

✓ Background Training with nohup

✓ Comprehensive Monitoring System

MAE Implementation

HYBRID RECONSTRUCTION INNOVATION

STANDARD MAE OUTPUT

- Shows ALL patches as reconstructed
- Visible patches look "garbled"
- Hard to assess reconstruction quality
- Misleading visual representation

**Why do visible patches look bad?
They're reconstructed, not copied!**

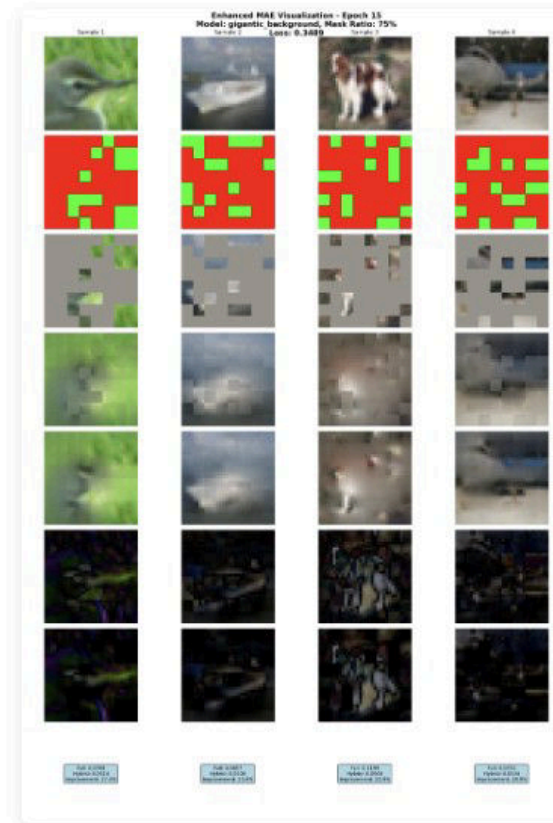
OUR HYBRID APPROACH

- Visible patches: Original image data
- Masked patches: Model predictions
- Clear quality assessment
- Accurate representation

**True reconstruction quality:
Only judge the predicted patches!**

ENHANCED TRAINING VISUALIZATIONS

GIGANTIC MODEL - ADVANCED RECONSTRUCTION ANALYSIS



ATTENTION PATTERNS

Visualize which patches the model focuses on

ACTIVATION DISTRIBUTIONS

Monitor neuron behavior across layers

WEIGHT EVOLUTION

Track parameter changes during training

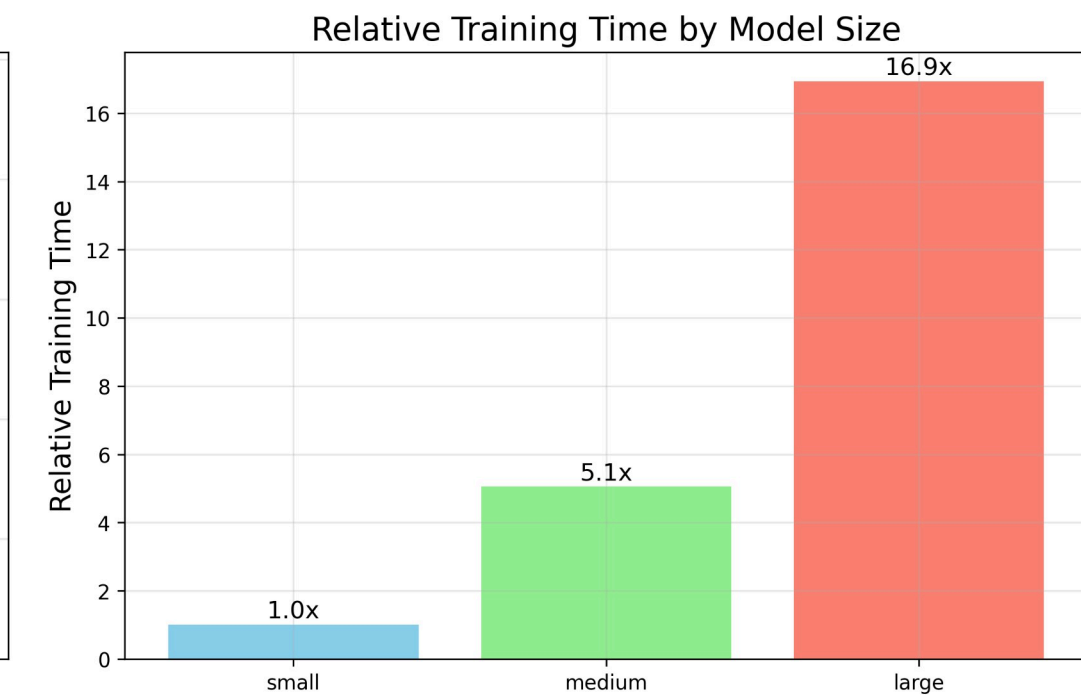
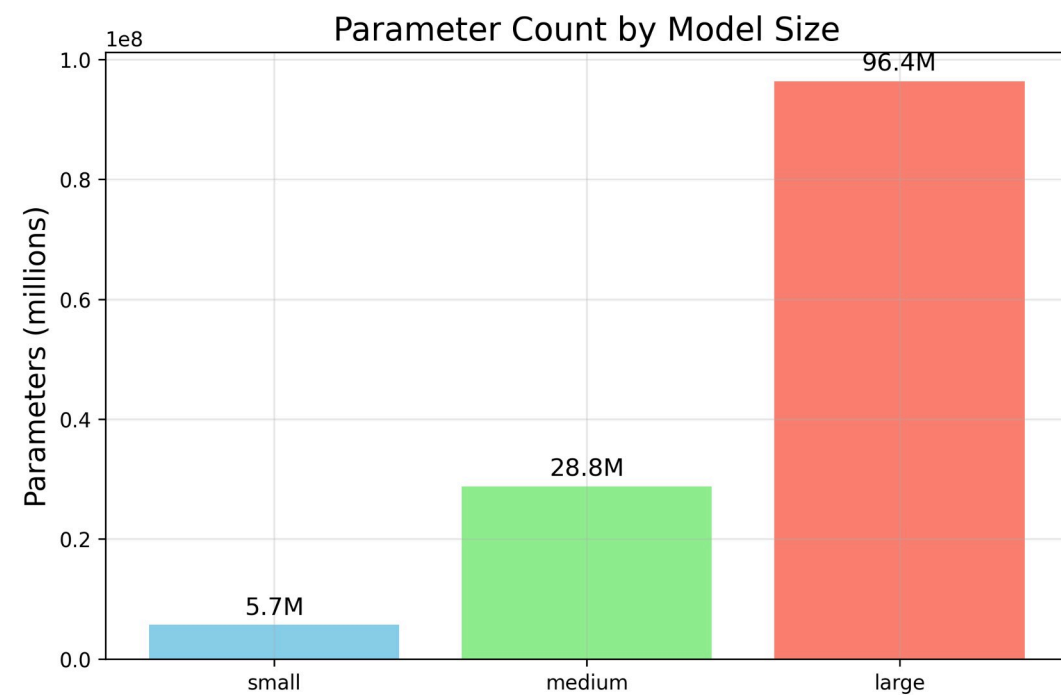
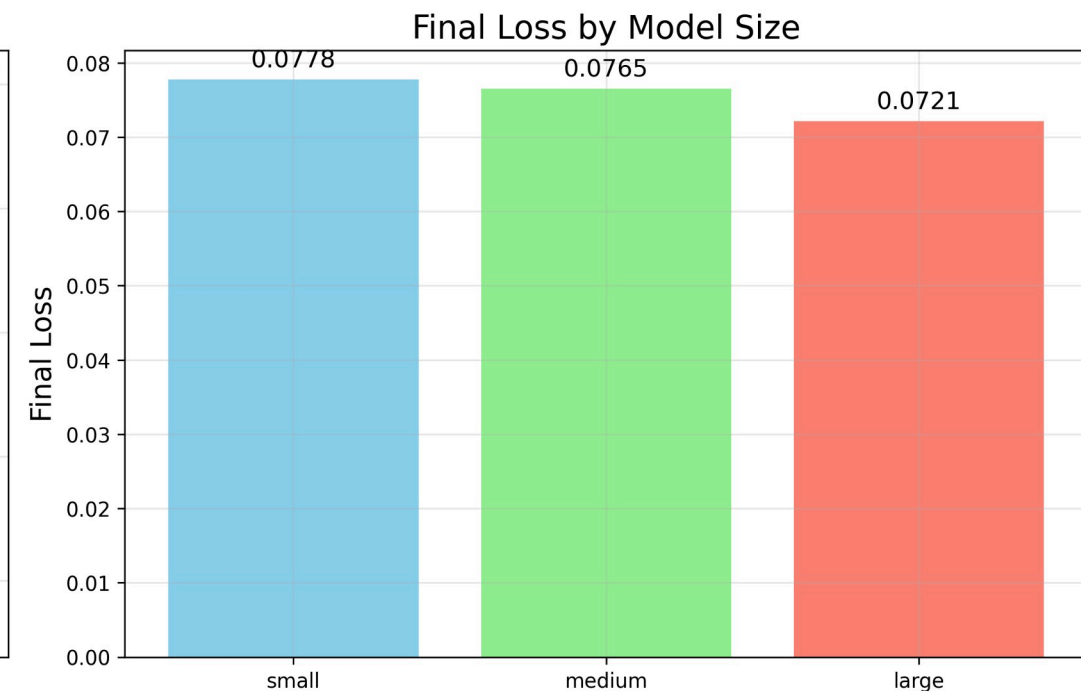
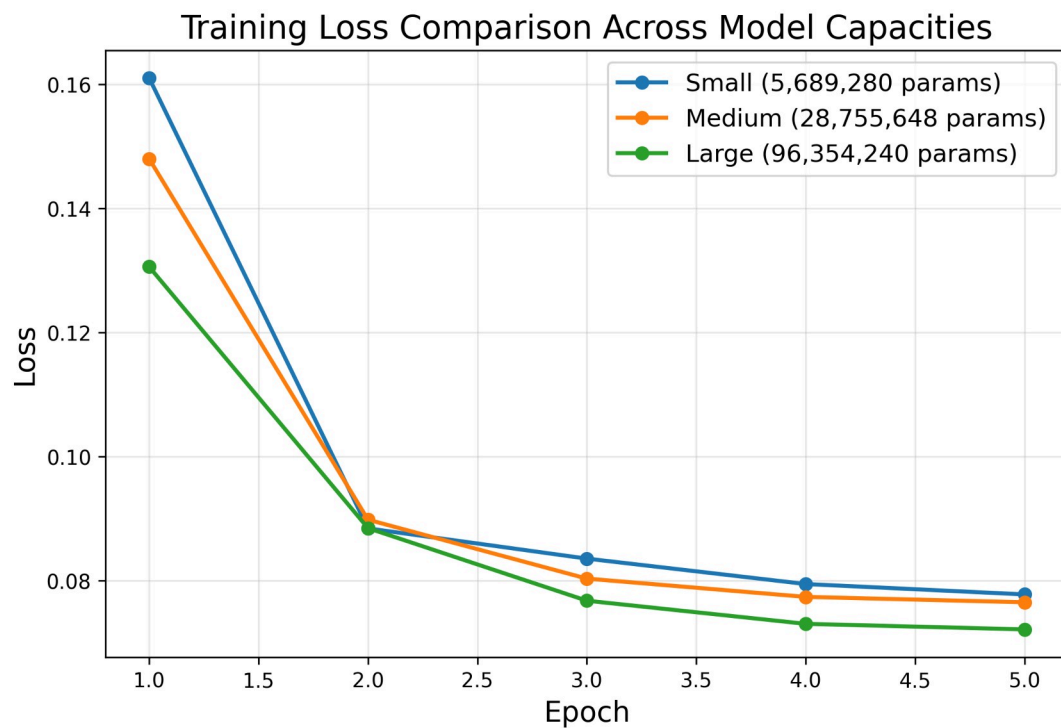
3D EMBEDDINGS

PCA visualization of learned representations

MAE Implementation

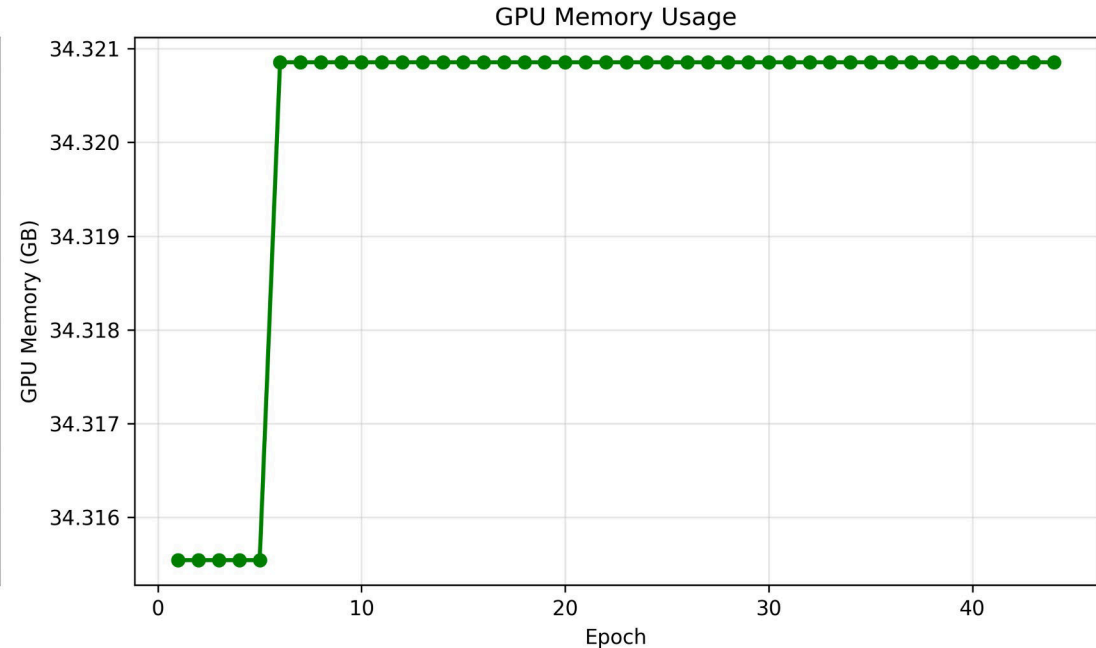
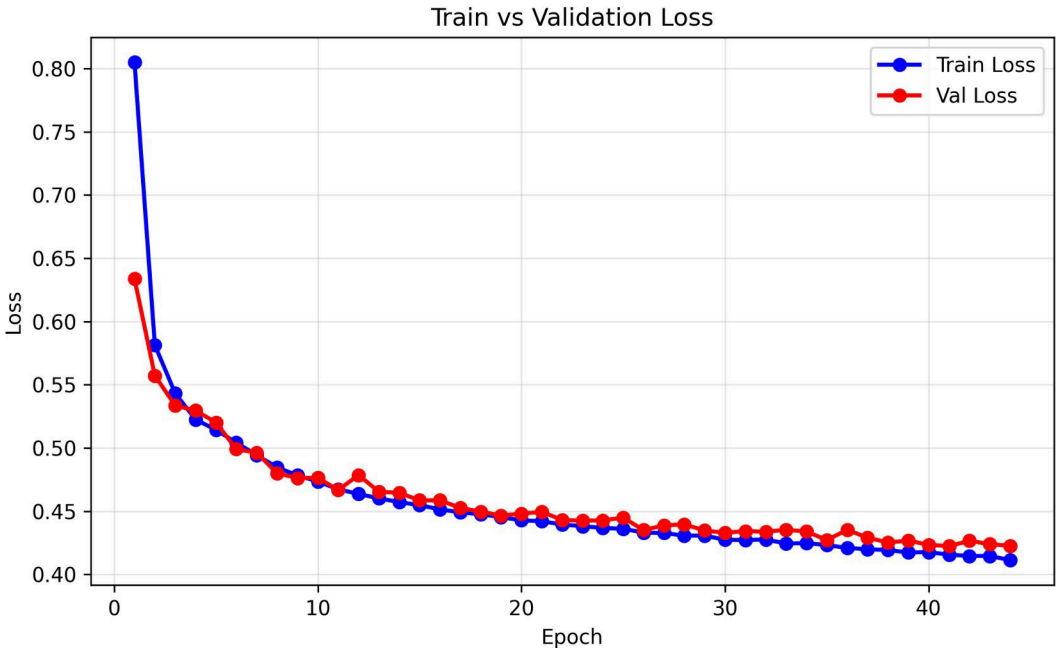
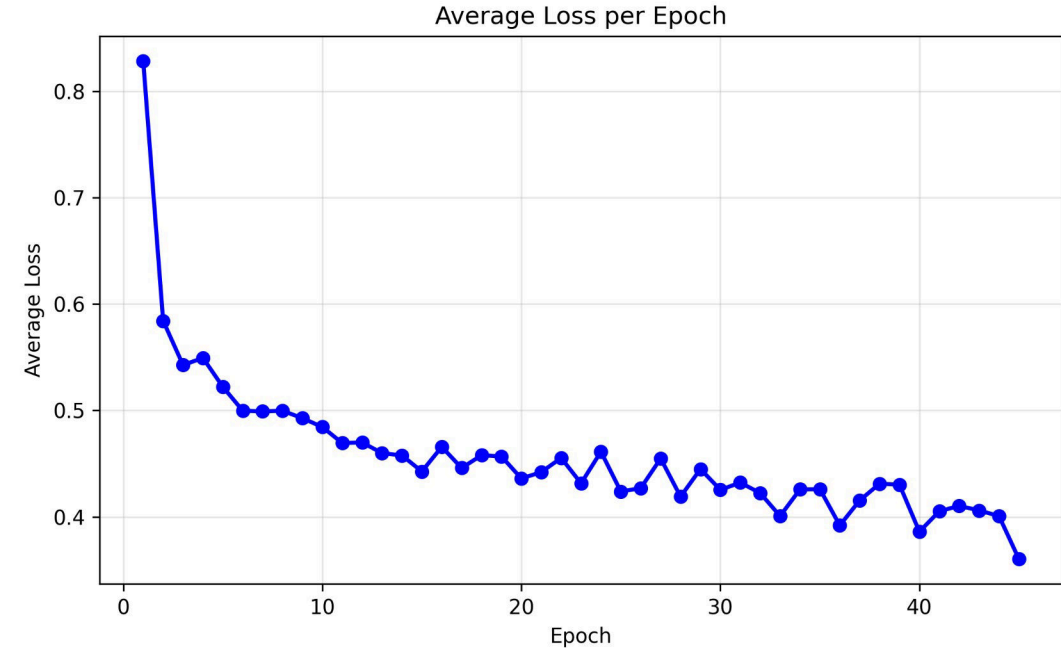
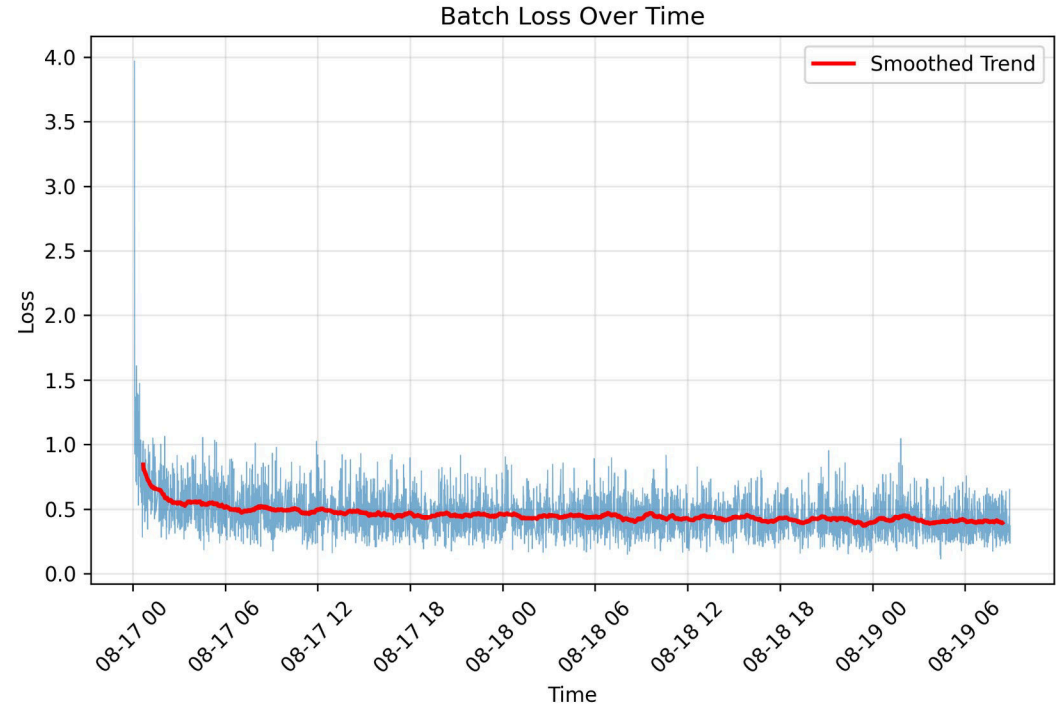
LESSONS LEARNED & FUTURE DIRECTIONS

-  **MASKING STRATEGY MATTERS MORE THAN EXPECTED** ^{*sometimes}
The standard 75% masking is not optimal for all scenarios - lower ratios ~~often~~ work better
-  **SCALING LAWS ARE COMPLEX**
Larger models help, but with diminishing returns - 2x parameters $\neq$ 2x better performance
-  **ADAPTIVE TRAINING IS ESSENTIAL**
Learning rate scheduling and early stopping significantly improve efficiency and results
-  **VISUALIZATION DRIVES UNDERSTANDING**
Hybrid reconstructions and neural visualizations provide crucial insights into model behavior



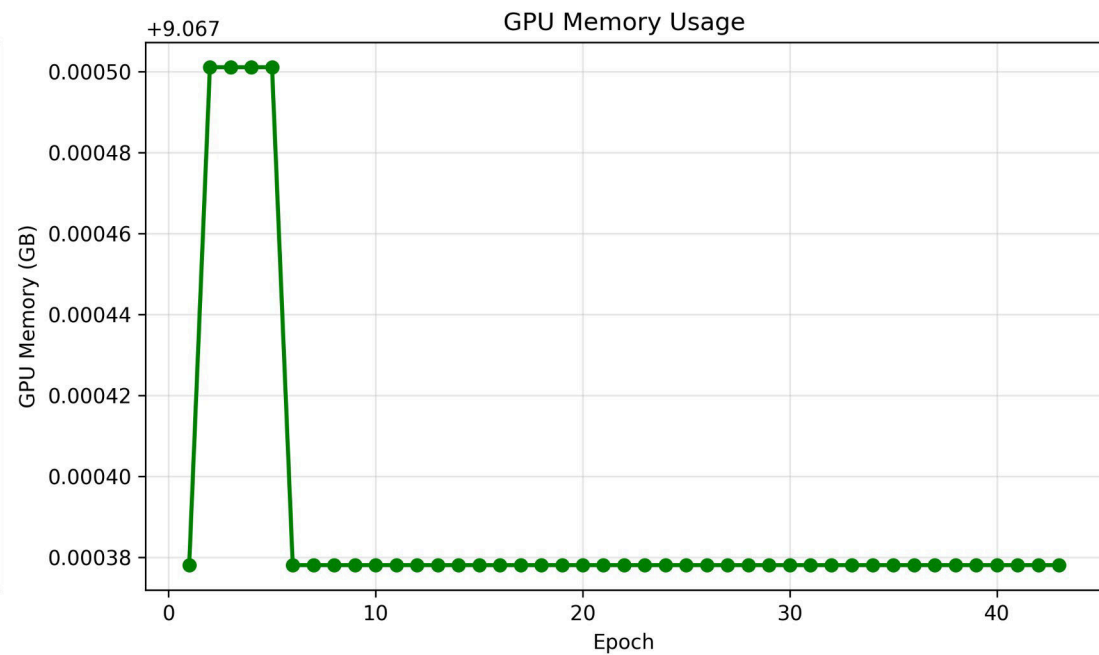
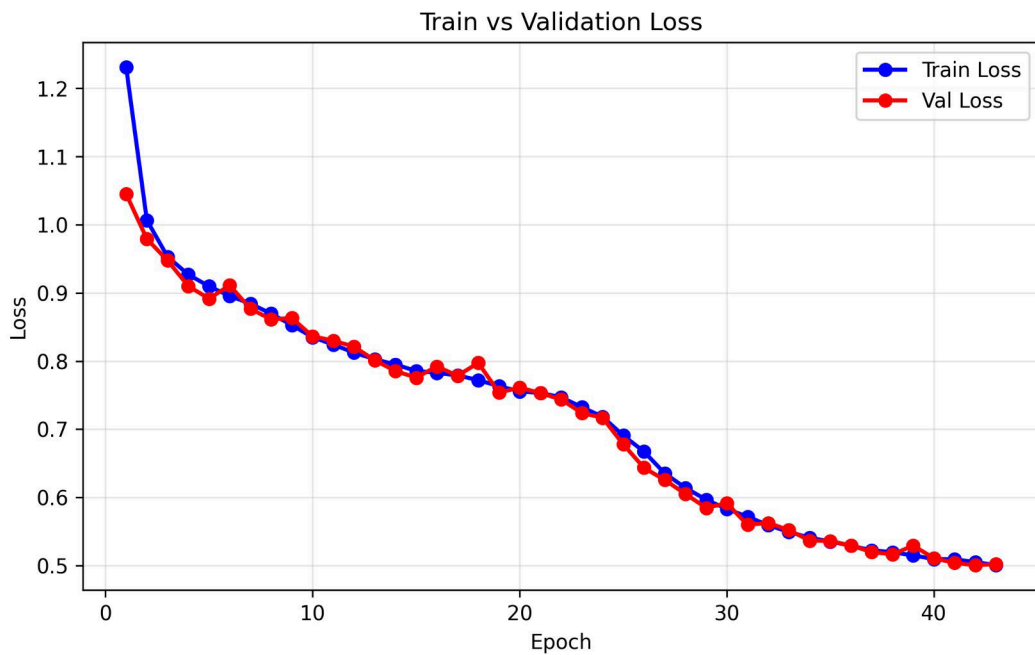
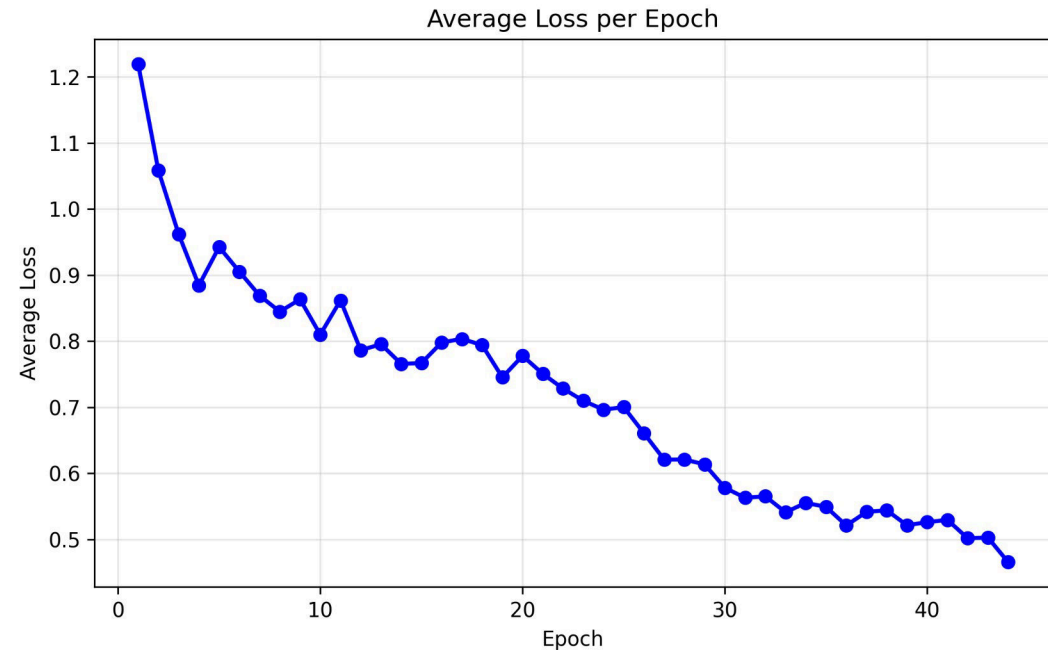
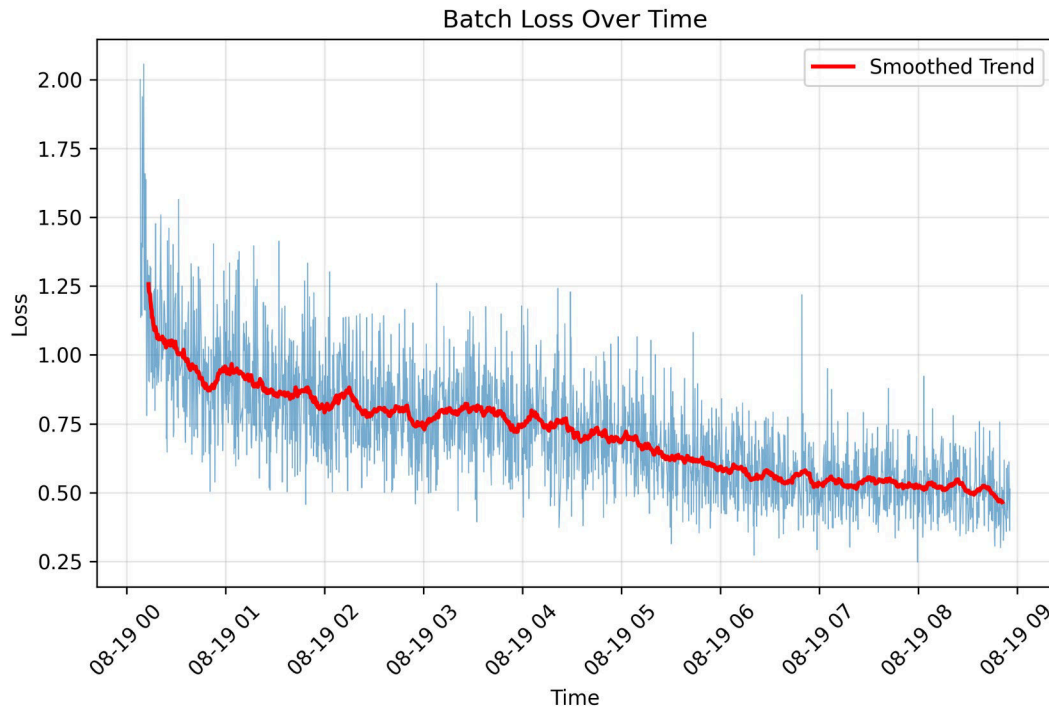
2.3B

Training Progress - Gigantic Background Experiment



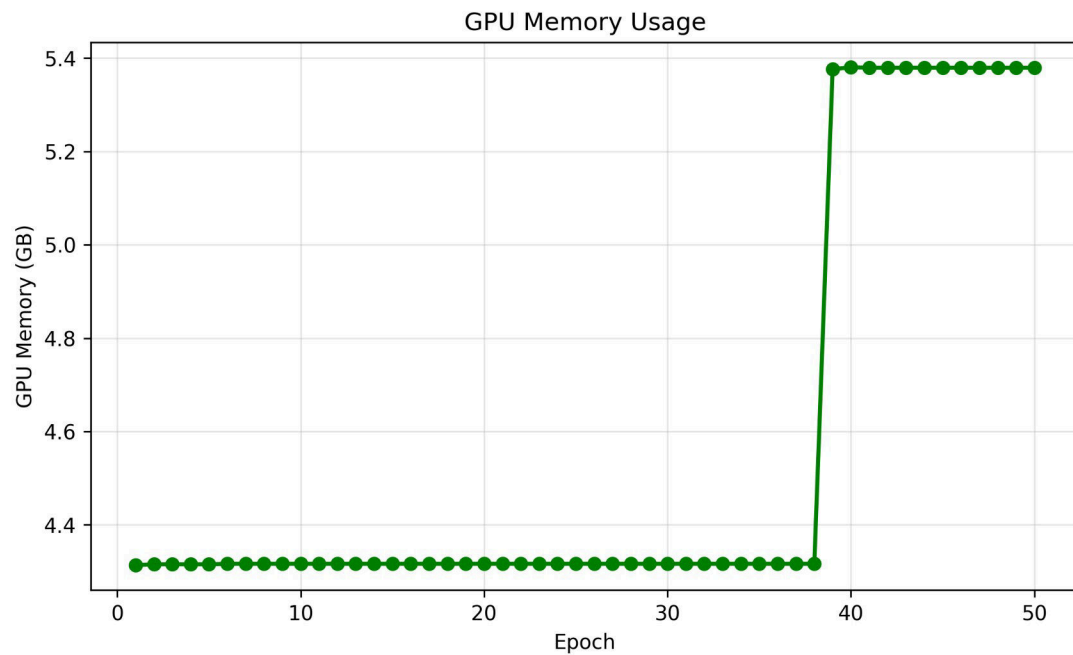
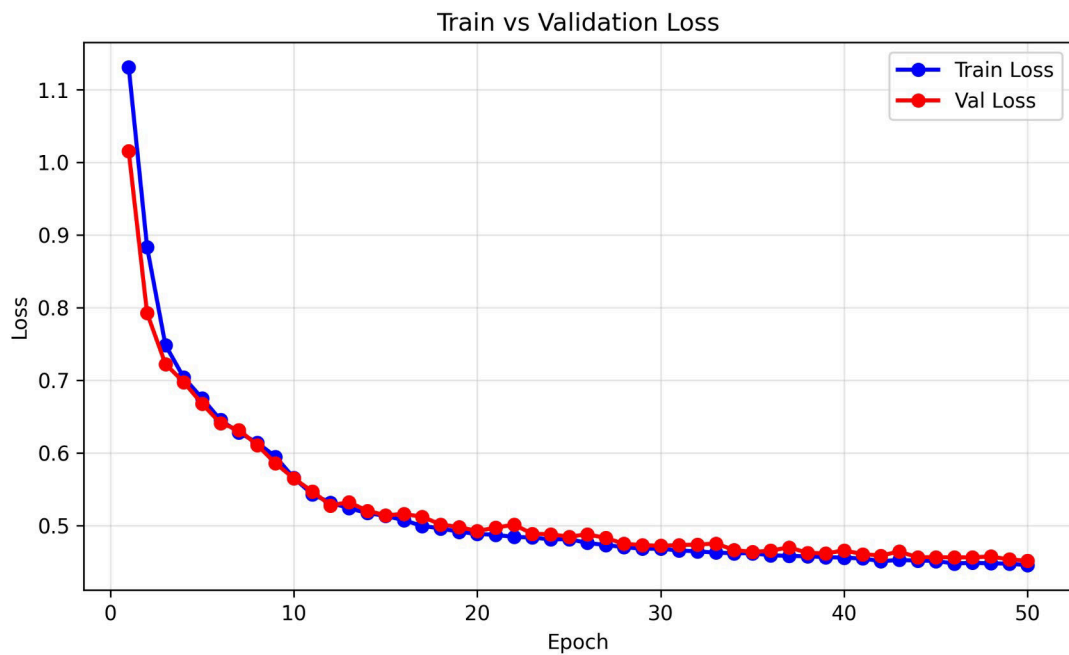
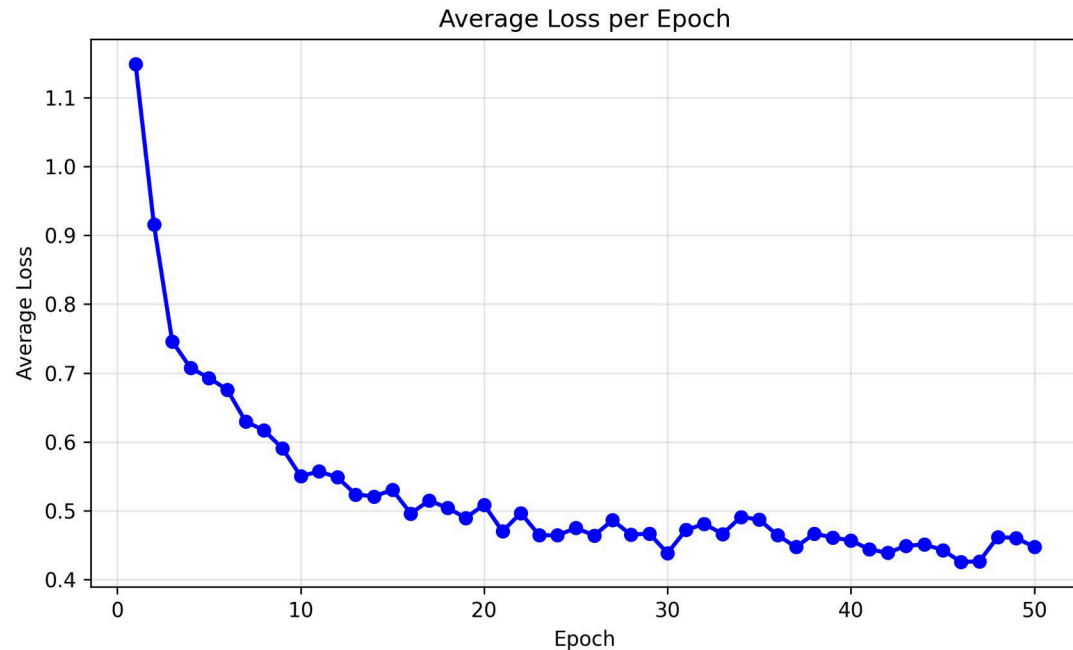
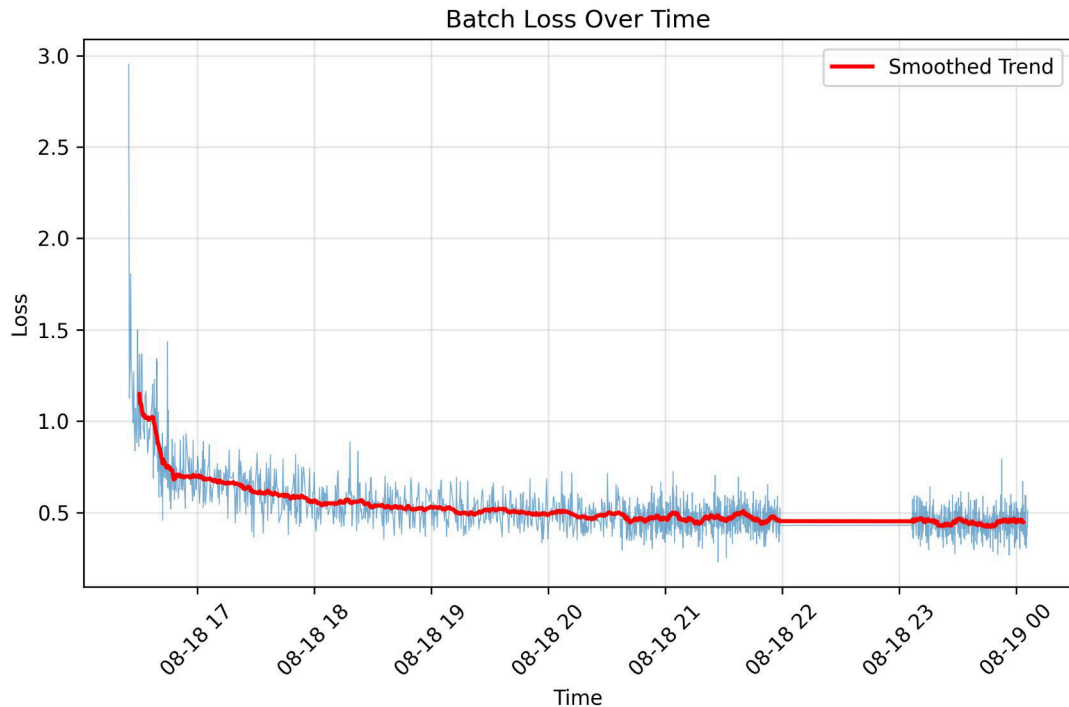
XXL

Training Progress - Gigantic Background Experiment



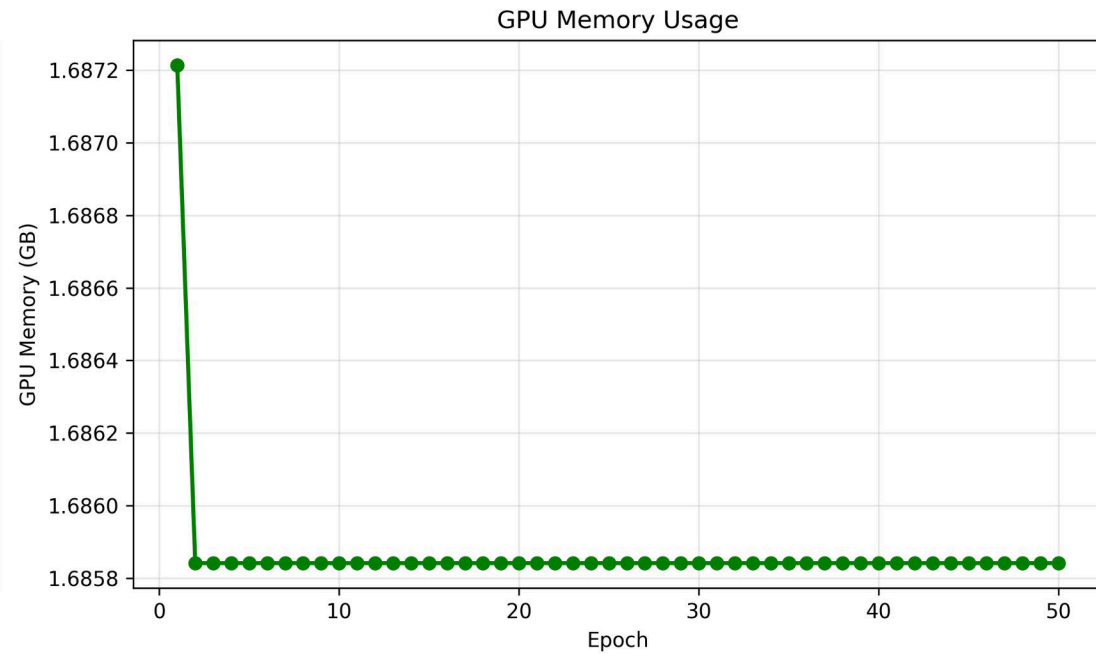
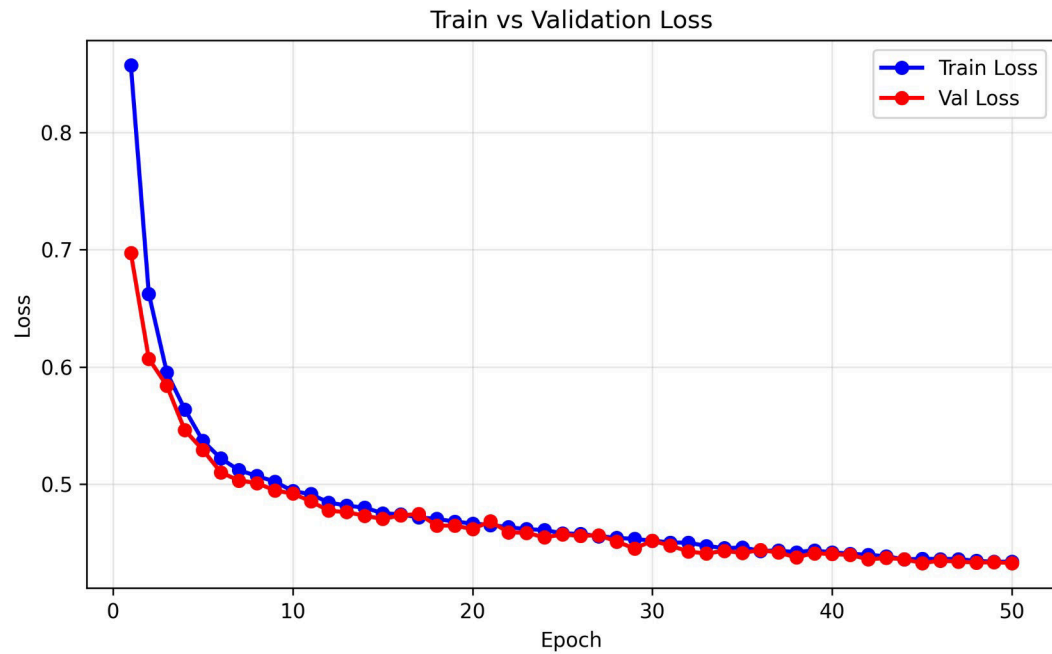
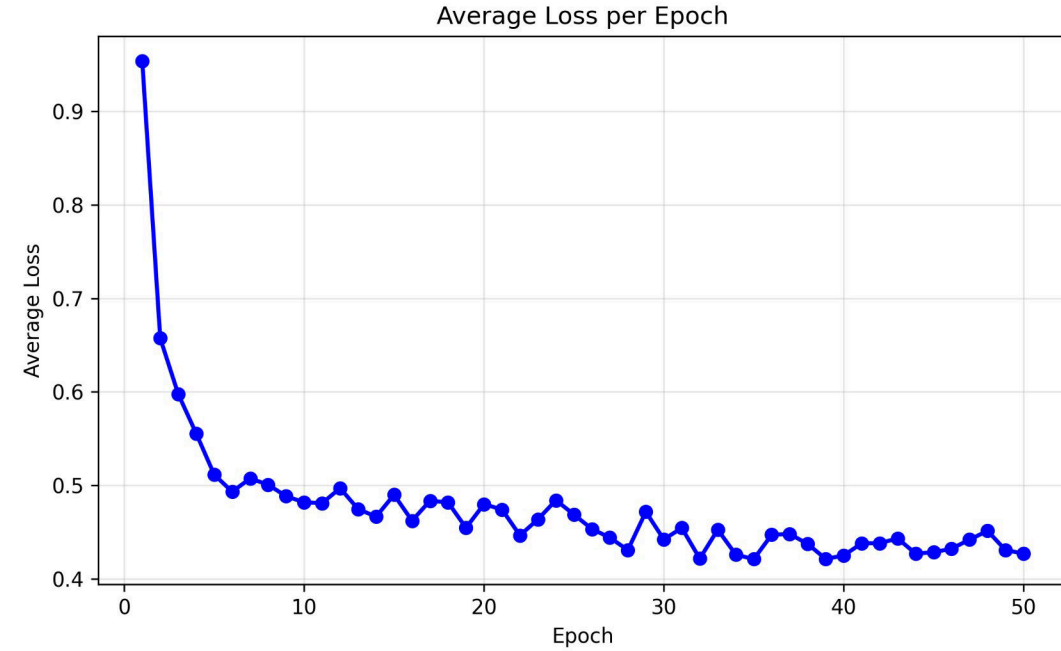
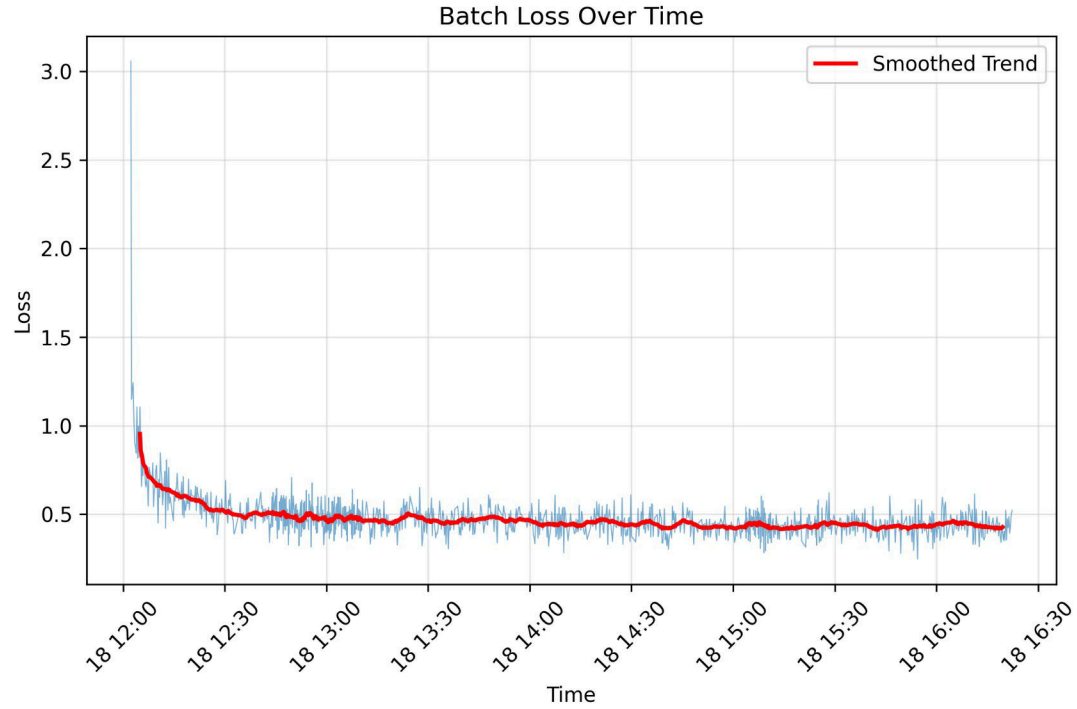
XL

Training Progress - Gigantic Background Experiment



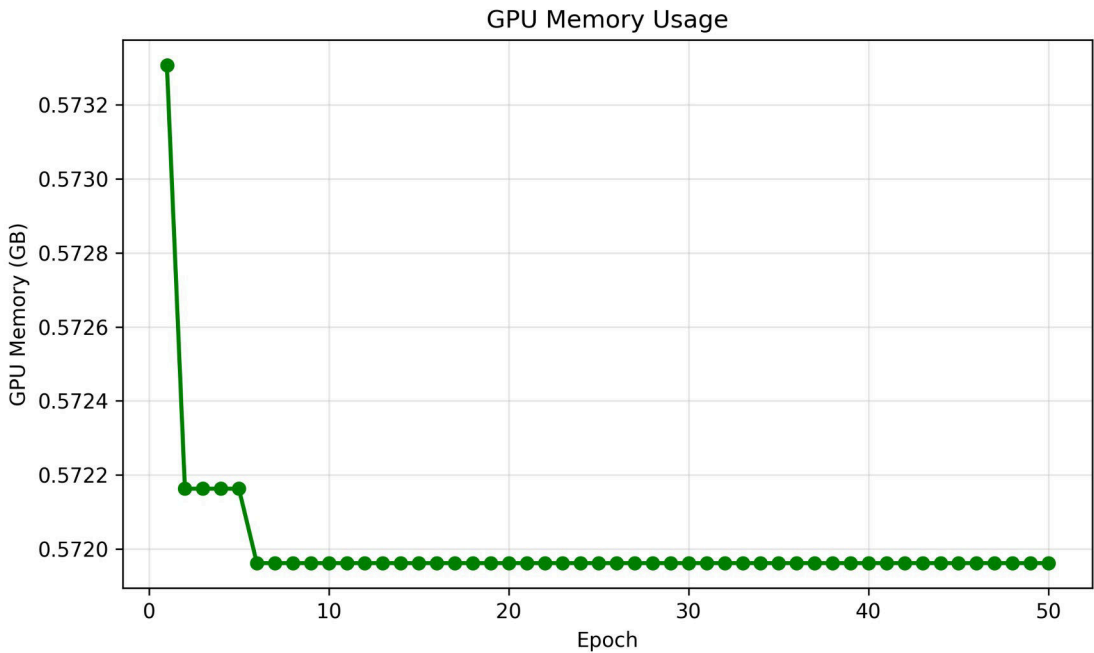
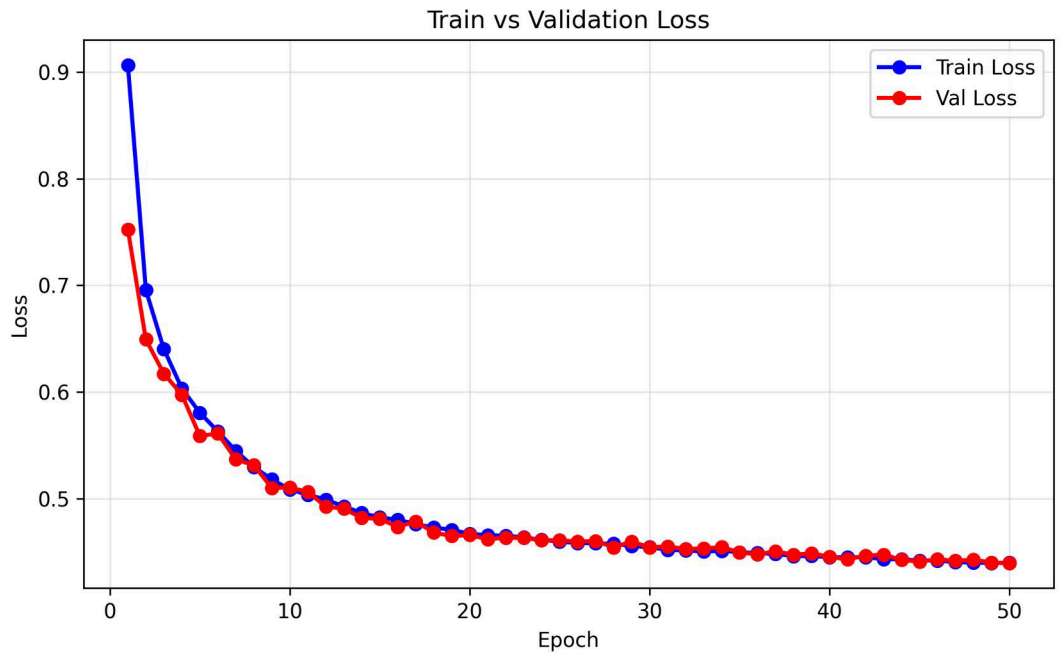
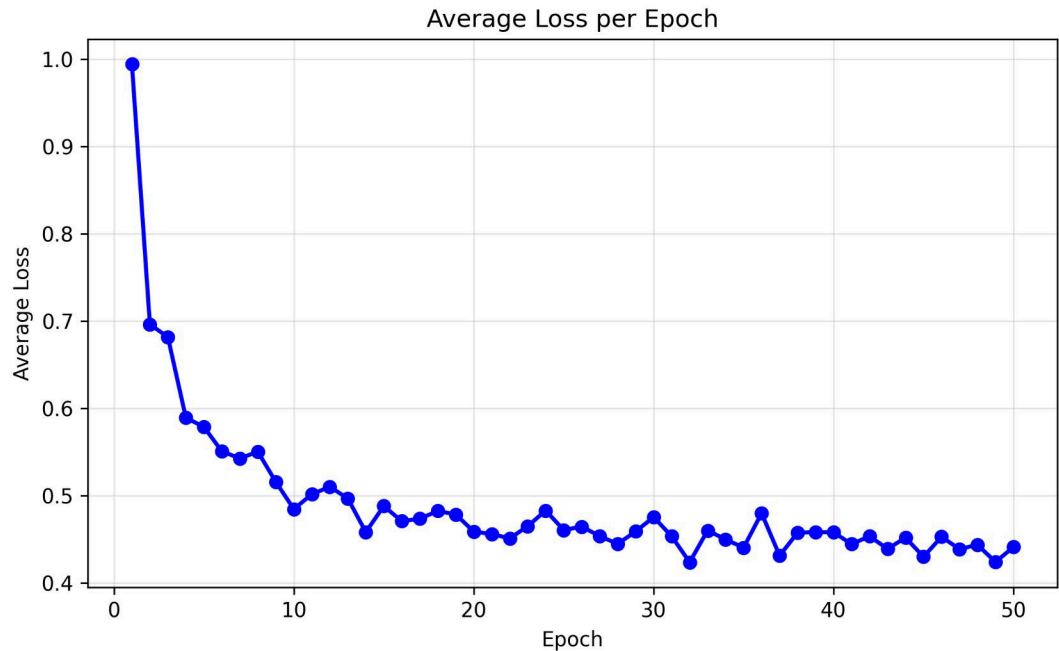
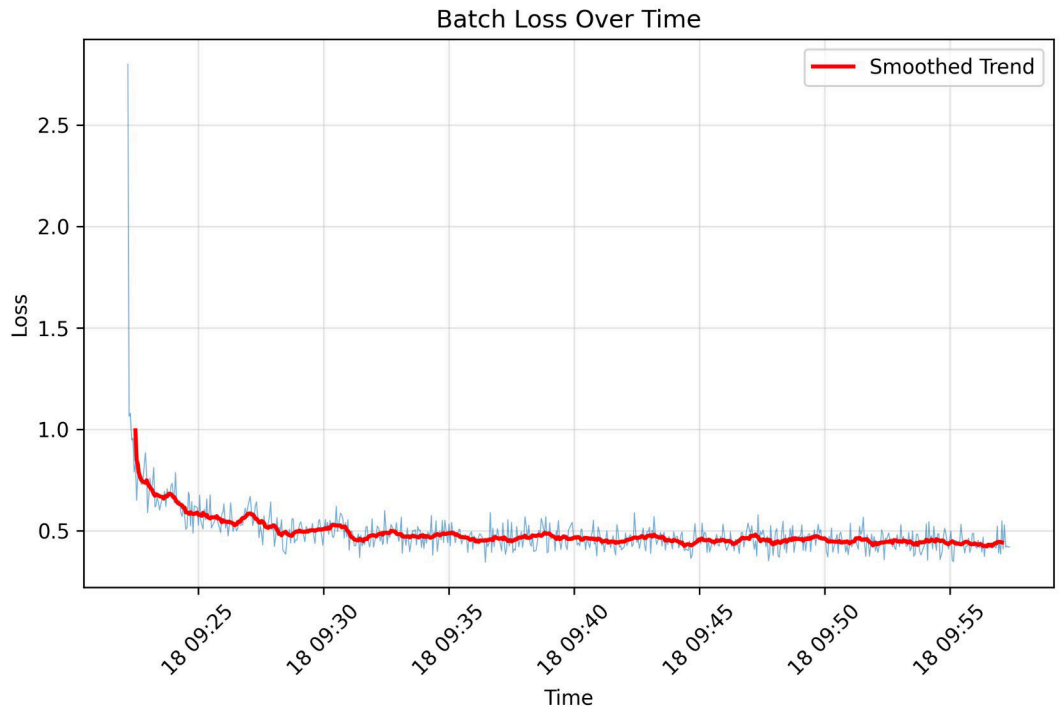
L

Training Progress - Gigantic Background Experiment



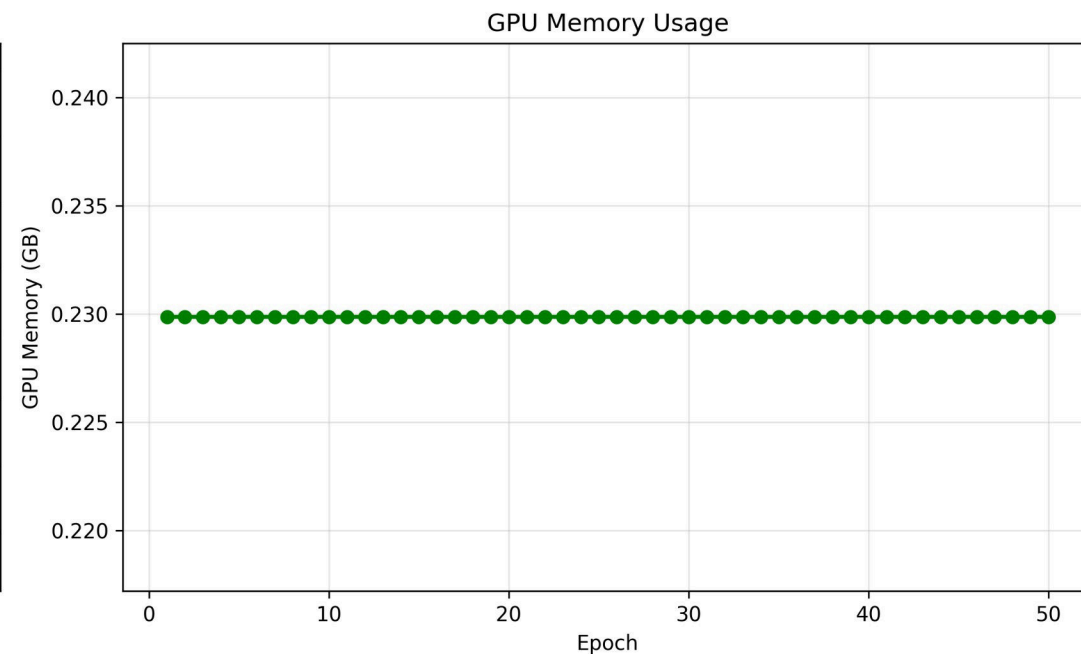
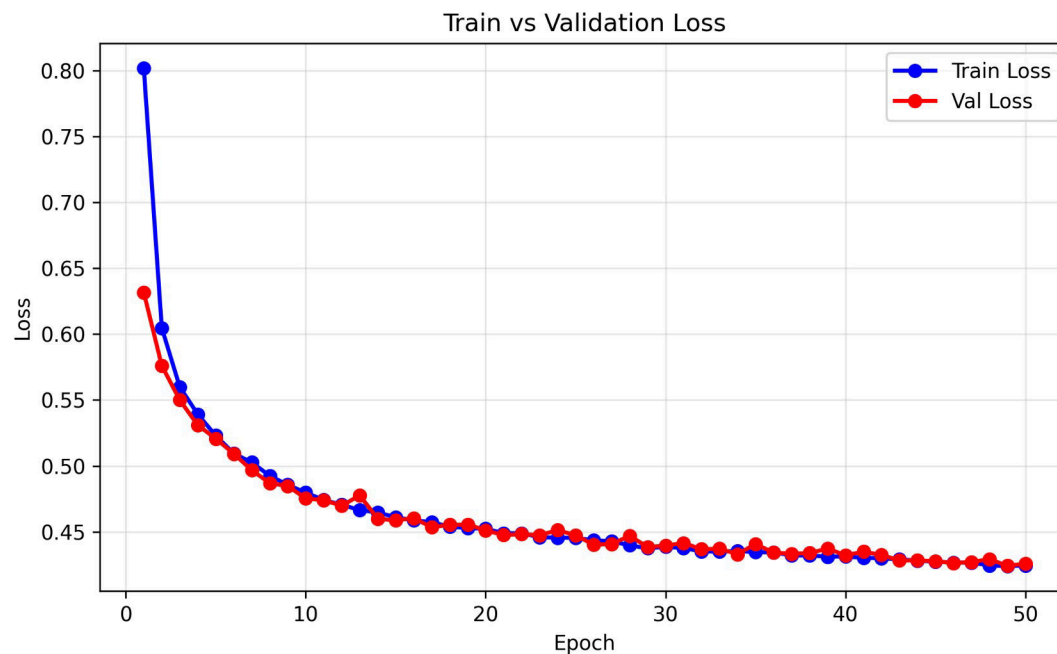
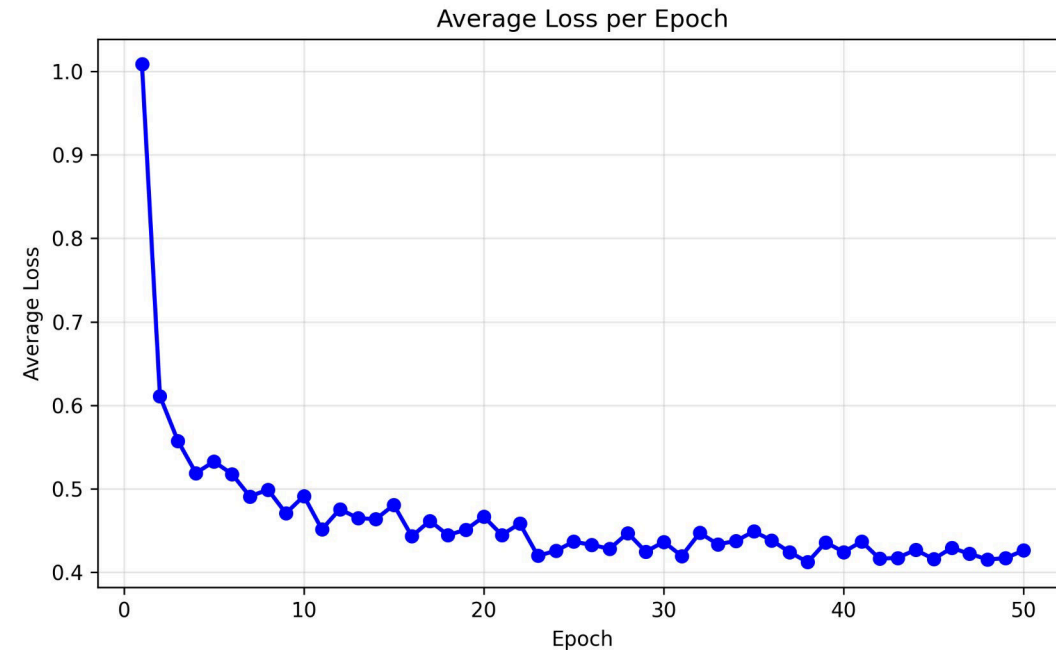
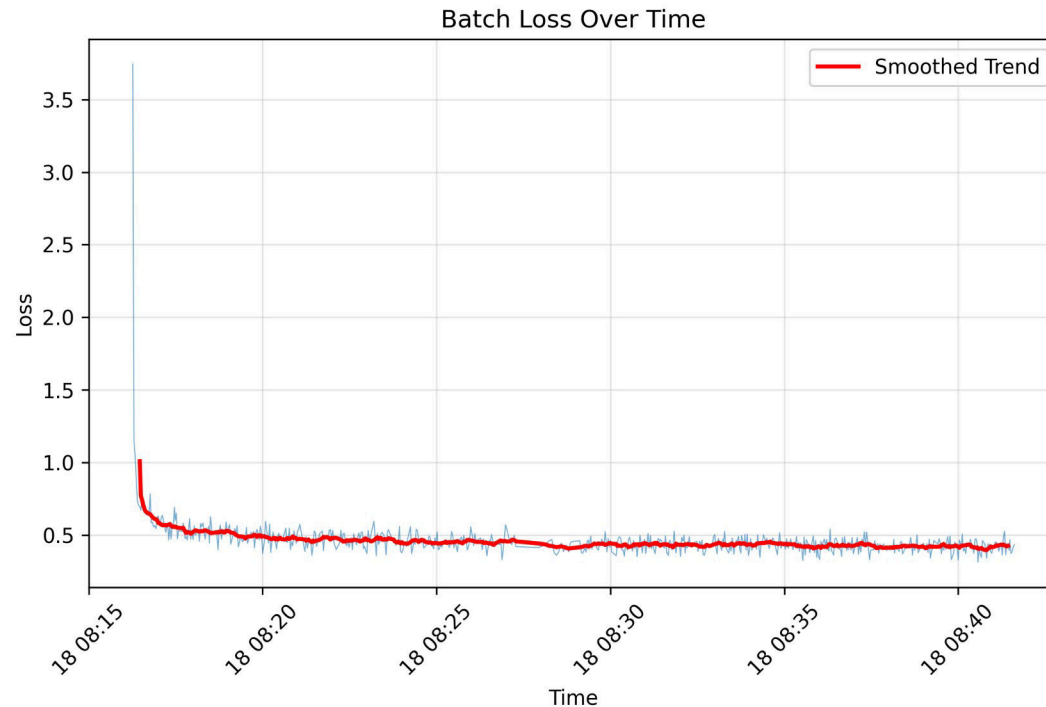
M

Training Progress - Gigantic Background Experiment



S

Training Progress - Gigantic Background Experiment

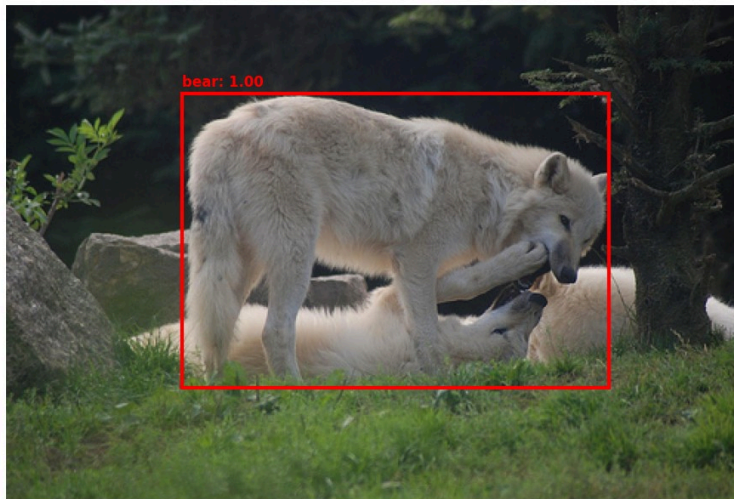


Zero Shot Segmentation

Original Image



Object Detection (Bounding Boxes)



Segmentation Masks (Basic)



Original Image



First Mask (Raw Values)
Min: 0.000, Max: 1.000



First Mask (Thresholded)
Non-zero pixels: 40340



MAE Impl

Original Image



Segmentation Masks



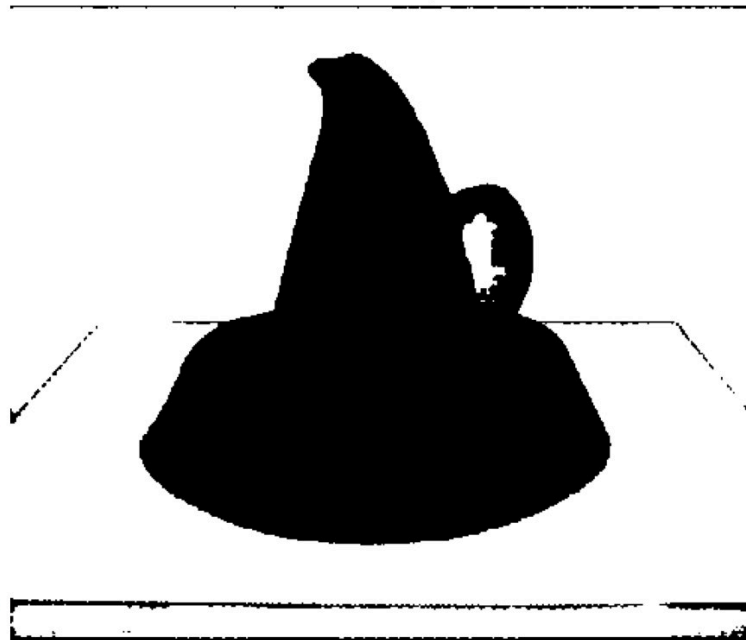
Labeled Segments



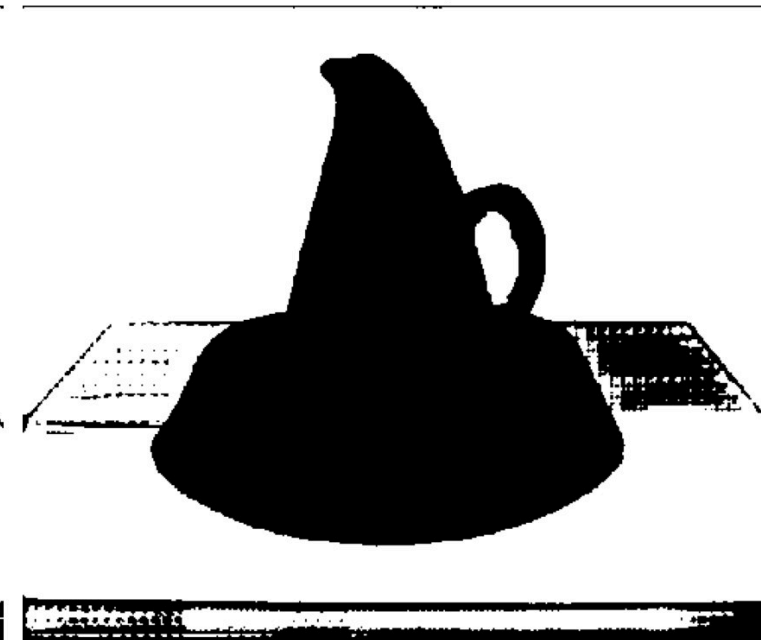
Original Image



First Segment Mask
Area: 149561 pixels



Second Segment Mask
Area: 140304 pixels



Original Image



Segmentation Masks



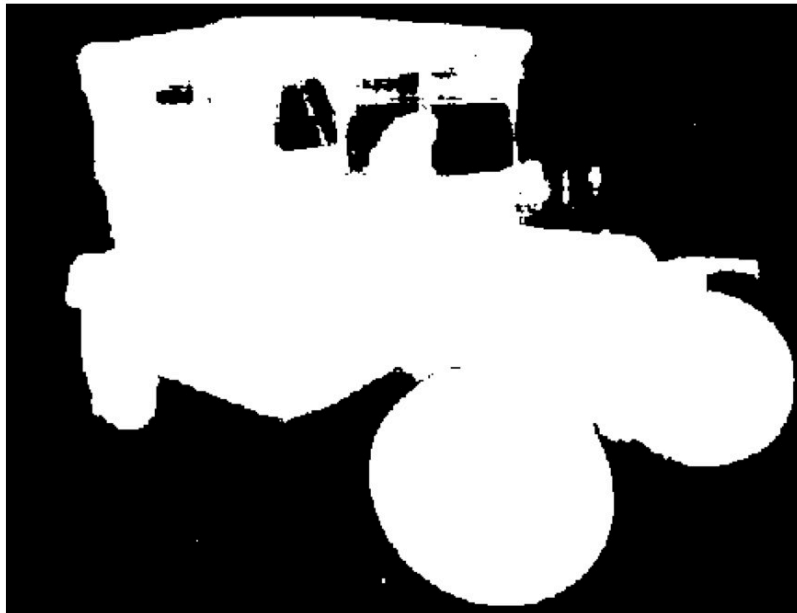
Labeled Segments



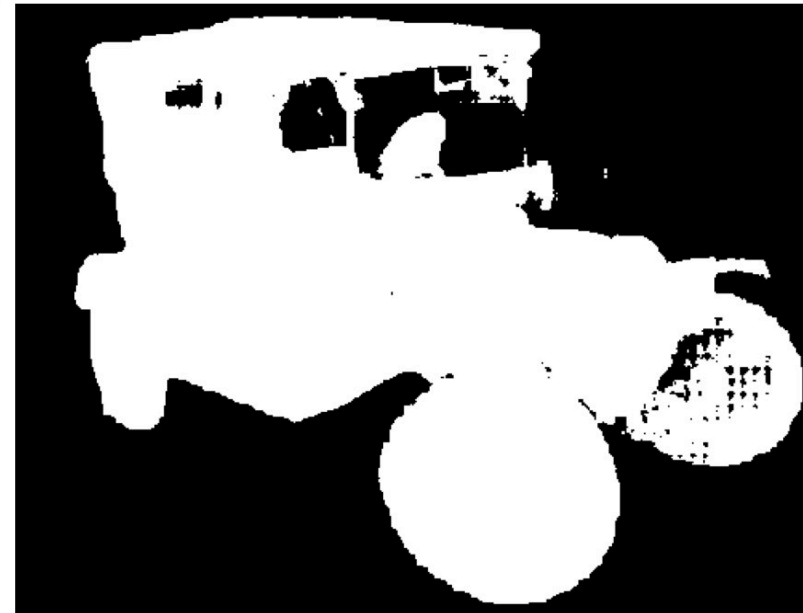
Original Image



First Segment Mask
Area: 94494 pixels



Second Segment Mask
Area: 90428 pixels



Testing ViT(Vision Transformers)

- Attempted to try a zero-shot segmentation algorithm
 - Used OpenAI CLIP
 - Grounding DINO
 - Meta's SAM
- Too much effort for something that might not pan out – could be future work